Optical Determination of Space-Debris Spin State

A ground-based, two-axis photometric testbed for recovering the rotation rate, spin-axis orientation and coarse shape of an uncooperative orbiting body from its reflected light curve.

DOCUMENT	SDC-001 — Design & Build Specification
REVISION	A — issued at design freeze
BUILD WINDOW	11 days, procurement to demonstration
TOTAL BUILD COST	See Section 11. Full build INR 12,718; lean variant INR 9,098
CAMERA	iPhone, 60 fps, fully manual exposure lock
PRIMARY DELIVERABLE	Recovered spin parameters validated against commanded ground truth
STATUS	Design frozen — released for procurement and build

WHAT THIS DOCUMENT IS

This is a complete build specification, not a concept note. Every component required to construct the instrument is enumerated in Section 11 with its part specification, quantity, unit price, function and procurement source — down to individual fasteners and consumables. Sections 6 to 9 give dimensioned drawings sufficient to build from. Section 12 gives the assembly sequence, Section 13 the commissioning procedure, and Section 15 an honest feasibility assessment including what this instrument cannot do.

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11	Bill of materials	Complete itemised parts list with costs
12	Assembly procedure	Ordered build sequence with checkpoints
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15	Feasibility assessment	Technical, schedule and cost realism; known limitations
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A	Pin assignment reference	Controller pin map and driver settings
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SECTION 1

What is being built

In one sentence: a bench instrument that spins a scale model of a satellite in a controlled two-axis tumble under a single light source, records the light it reflects with an ordinary phone camera, and runs software that reconstructs the tumble from that recording alone — then checks the reconstruction against the motion that was actually commanded.

The deliverable is not the mechanism. The deliverable is a **demonstrated inversion**: a number for rotation period, spin-axis orientation and panel area, each with an error bar, each compared against a value that is independently known. The mechanism exists to generate a signal whose true answer is known in advance, which is the only way to prove that the analysis works.

1.1 Why a laboratory instrument rather than a telescope

The scientific target is real orbiting debris. A ground telescope observing an actual derelict produces a light curve, and software can produce a spin period from it — but nobody knows the true spin period of that object, so there is no way to grade the answer. The claim would be unfalsifiable.

Building a target whose motion is commanded removes that problem entirely. The rotation rate is set in firmware, the spin axis geometry is set by the mechanical design, and an independent optical sensor measures the actual revolution period without reference to the camera at all. The inversion can therefore be scored, and its error bars can be verified rather than asserted.

> **THE DESIGN PRINCIPLE THAT GOVERNS EVERY CHOICE BELOW**
> Every subsystem in this build exists either to **produce a known signal** or to **measure that signal without contaminating it**. Where a design decision trades convenience against the integrity of the measurement, this document takes the measurement. That is why the illumination is DC rather than mains-driven, why the stepper drivers are TMC2209 rather than the cheaper A4988, why the camera exposure is locked manually, and why an eighty-rupee slot sensor is treated as a critical component.

1.2 What the finished instrument does

— Rotates a printed target continuously about one axis while rocking that axis through a second, producing a light curve with two incommensurate frequencies — the signature of genuine non-principal-axis tumbling rather than a trivial single-frequency spin.

— Illuminates the target with a single collimated, flicker-free source at a realistic 30° phase angle, inside a light-sealed enclosure.

— Records reflected flux at 60 frames per second with all camera automation defeated.

— Extracts a calibrated brightness time series, searches it for periodicity, and fits a physical reflection model to recover spin rate, axis orientation, cone angle and approximate panel area.

— Reports every recovered parameter against commanded truth and against the independent slot-sensor period, with residuals and uncertainties.

1.3 What it explicitly does not do

Stating this plainly is part of the specification. An instrument whose limits are declared is more credible than one whose limits are discovered by an examiner.

— It does not observe real orbital targets. Atmospheric seeing, target faintness, short pass durations and tracking are outside this build.

— It does not produce absolute radiometry. The camera response is non-linear and only approximately linearised; recovered areas are relative, not calibrated in physical units.

— It does not implement full convex light-curve inversion of the kind used in asteroid shape modelling. The forward model here is a parameterised box-wing, which is appropriate for artificial satellites but is not a general shape solver.

SECTION 2

Why spin state matters

Active debris removal is the operational driver. A servicer spacecraft that intends to capture a derelict — by robotic arm, net, harpoon or docking probe — must know how that derelict is moving before it arrives. The consequences of not knowing are concrete:

— **Matching the tumble costs propellant** and may exceed the servicer's own attitude control authority.

— **Capturing without matching transfers angular momentum.** If the combined body's momentum exceeds what the servicer's wheels and thrusters can dump, the result is two tumbling objects instead of one.

— **The capture mechanism must be designed around the motion** — which means the spin state is needed years before the mission flies, not on arrival.

— **A failed capture can fragment the target**, producing thousands of new objects. That is the precise opposite of the mission's purpose.

2.1 The economic argument

The alternative to characterising a target from the ground is to fly a mission to inspect it. Astroscale's ADRAS-J did exactly this in 2024, rendezvousing with a spent Japanese upper stage and imaging it largely to establish how it was moving. That is a mission-scale answer to a single question about a single object.

Ground-based light-curve inversion costs effectively nothing per target. The value proposition is therefore **pre-screening**: characterise many objects cheaply, then commit expensive flight hardware only to those whose motion is understood well enough to approach safely. Thousands of large derelicts are tracked positionally; only a small fraction have well-determined spin states.

2.2 Spin states are not static

Rotation rates evolve through eddy-current damping against the Earth's magnetic field, internal energy dissipation, solar radiation pressure torque and gravity-gradient effects. A characterisation from several years ago is stale. This makes the problem a monitoring task rather than a one-time survey — which is precisely the kind of task a cheap, replicable ground instrument suits.

2.3 Four further applications

Application	How the light curve is used
Anomaly diagnosis	A satellite goes silent. The light curve distinguishes a transmitter failure from a total attitude-control loss within a single pass. Operators use this routinely.
Space domain awareness	An unexplained change in spin state indicates that something happened — a collision, a fragmentation, an outgassing event or a deliberate manoeuvre. Detecting the change has value independent of explaining it.
Reentry prediction	A tumbling body presents a different mean cross-section than a stabilised one, changing its drag and therefore its decay rate and reentry footprint.
Object identification	Shape and spin signature help separate a rocket body from a payload from a fragment where the catalogue entry is ambiguous.

THE FRAMING TO USE

Stabilising the low-Earth-orbit environment is commonly assessed to require the removal of roughly five to ten large objects per year. Every one of those removals requires the target's spin state to be known before anything can touch it. This instrument addresses the cheap prerequisite to an expensive problem that is already widely agreed to be urgent.

SECTION 3

Measurement principle

A derelict satellite emits no light of its own. It is a lump of metal and composite reflecting sunlight, exactly as the Moon does. Because it is not spherical — solar panels, a boxy bus, antennas — the fraction of incident light it directs toward an observer depends on its instantaneous orientation.

As the body rotates, that fraction varies. A flat solar panel turning face-on to the observer produces a sharp specular glint; the same panel edge-on returns almost nothing. Recording brightness against time therefore produces a periodic trace — the **light curve** — whose rhythm, harmonic content and profile shape encode the rotation rate, the orientation of the spin axis, and coarse information about the reflecting geometry.

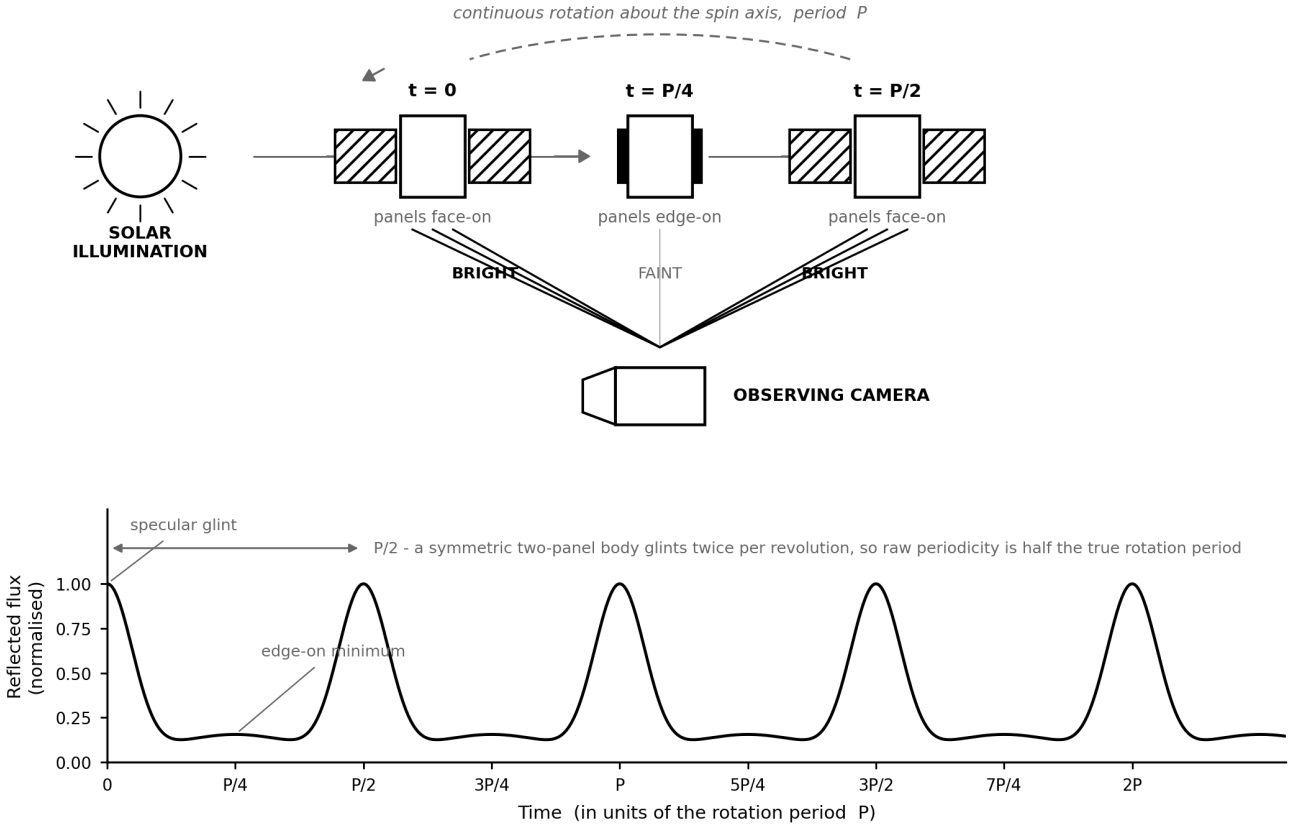


FIG. 1 — Measurement principle. A rotating two-panel body modulates the flux reaching the observer. Note that a symmetric body glints twice per revolution, so the dominant periodicity in the raw signal is P/2 rather than the true rotation period P. Resolving this ambiguity is a required step in the analysis, not an optional refinement.

3.1 The two reflection components

The forward model used in Section 10 treats each surface as the sum of two terms, because real spacecraft surfaces behave in both ways and the contrast between them is what makes the light curve informative:

Component	Physical origin	Signature in the light curve
Diffuse (Lambertian)	Multilayer insulation, painted structure, composite panels. Scatters incident light broadly.	Slow, smooth, low-amplitude modulation. Sets the baseline level.
Specular	Solar cell cover glass, polished metal, flat radiators. Mirror-like, strongly directional.	Narrow high-amplitude spikes at specific orientations. Carries most of the geometric information.

This is why Section 9 insists that the printed target be surfaced. Bare PLA is uniformly diffuse in every direction; a target printed and left untreated produces a weak, featureless curve with little to invert. The aluminised film and the blue transparent facing exist to create the specular component.

3.2 What is recoverable, and what is not

Quantity	Recoverable?	Basis
Rotation period	Yes, precisely	Dominant frequency of the periodogram once harmonic ambiguity is resolved.
Second (precession) period	Yes	Envelope modulation frequency, well separated from the spin frequency by design.
Spin-axis cone angle	Yes, moderately	Controls the depth and asymmetry of the envelope.
Relative panel area	Approximately	Amplitude of the specular component relative to the diffuse baseline.
Absolute size	No	Degenerate with distance and albedo from photometry alone.
Spin-axis direction in an inertial frame	Not in this build	Requires observations from multiple geometries over time.

SECTION 4

System architecture

The instrument divides into three subsystems with a single signal path between them. The division matters because each can be built and tested independently, which is what makes an eleven-day schedule survivable.

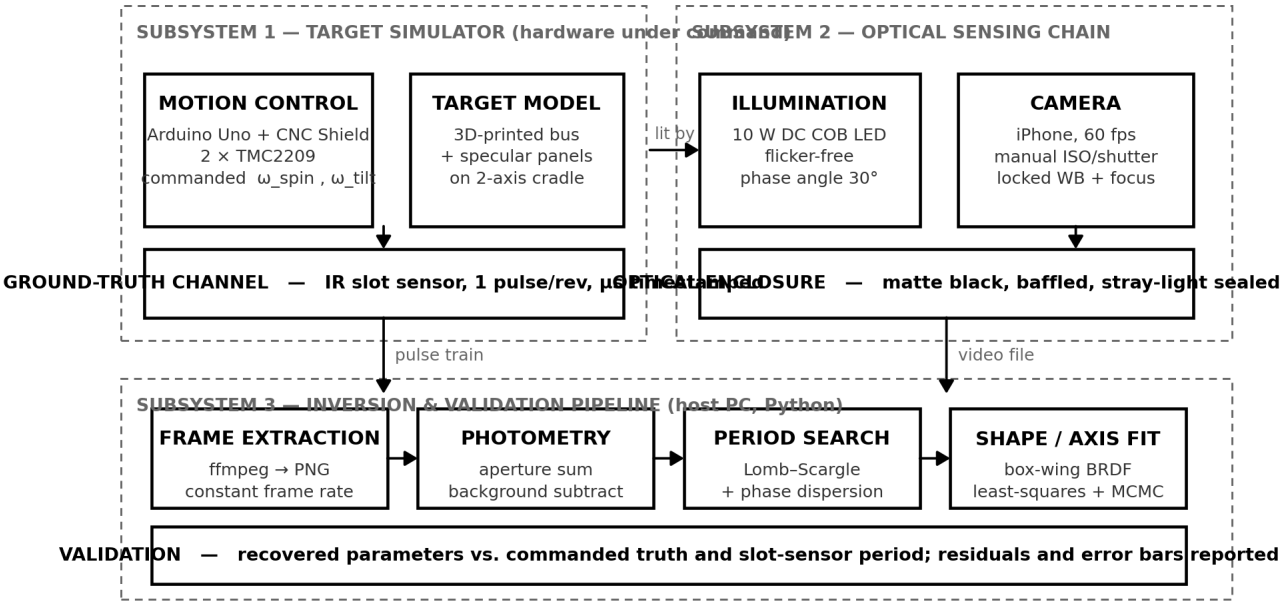


FIG. 2 — System architecture. Subsystem 1 generates a commanded tumble and reports its true rotation period through an independent channel. Subsystem 2 converts reflected light into a recorded video file. Subsystem 3 reduces that file to physical parameters and scores them against truth. The dashed ground-truth channel is what converts a demonstration into a measurement.

4.1 Subsystem responsibilities

Subsystem	Responsibility	Failure consequence
1 — Target simulator	Produce a repeatable, commanded two-axis tumble; report actual revolution timing independently of the optical chain.	Without it there is no known truth, and the inversion cannot be scored.
2 — Optical sensing	Convert reflected flux into a recorded intensity time series without introducing artefacts — no flicker, no auto-exposure, no stray light.	Contaminated input produces confident wrong answers, which is worse than no answer.
3 — Inversion pipeline	Reduce the recording to spin parameters with quantified uncertainty and compare against commanded truth.	The project's actual deliverable; the hardware exists only to feed it.

4.2 Interfaces

Interface	Medium	Specification
Host to controller	USB serial	115200 baud, 8N1. Command strings for rates; controller streams timestamped slot-sensor pulses.
Controller to drivers	STEP / DIR logic	5 V logic on shield headers. See Appendix A.
Drivers to motors	4-wire bipolar	Coil pairs identified by continuity before connection. Never hot-plugged.
Target to camera	Free-space optical	400 mm working distance, 30° phase angle, baffled against direct illumination.
Camera to host	File transfer	H.265 or ProRes .mov at constant frame rate; frame index divided by frame rate forms the time base.

TIMING — WHY NO GPS RECEIVER APPEARS IN THIS BUILD

An earlier concept for a field observing rig required a GPS receiver with a pulse-per-second output, because a satellite pass gives one chance and the host clock drifts by tens of milliseconds. In this laboratory configuration the **video frame rate is itself the clock**: frames are equally spaced by construction, so the time base is exact up to the accuracy of the camera's crystal. The slot sensor provides an independent absolute check. This removes an imported component with a multi-week lead time from the critical path — a decision forced by the eleven-day window and one that costs nothing in measurement quality.

SECTION 5

Kinematic design

A target that rotates about a single axis produces a clean, near-sinusoidal light curve. Any Fourier transform solves it in one line. Such a demonstration proves almost nothing, and a knowledgeable examiner will identify the weakness immediately.

Real derelicts do not behave that way. Energy dissipation drives a freely tumbling rigid body toward rotation about its axis of maximum inertia, but the approach takes years and most observed objects are caught mid-way, rotating about a non-principal axis. The observable consequence is a light curve containing **two incommensurate frequencies** that beat against each other, producing a signal whose period is not obvious by inspection.

This build reproduces that structure deliberately. Axis A rotates the target continuously. Axis B rocks the entire cradle carrying Axis A through ±60°, sweeping the spin axis through the observer's frame. The result is a genuinely difficult inversion problem at bench scale.

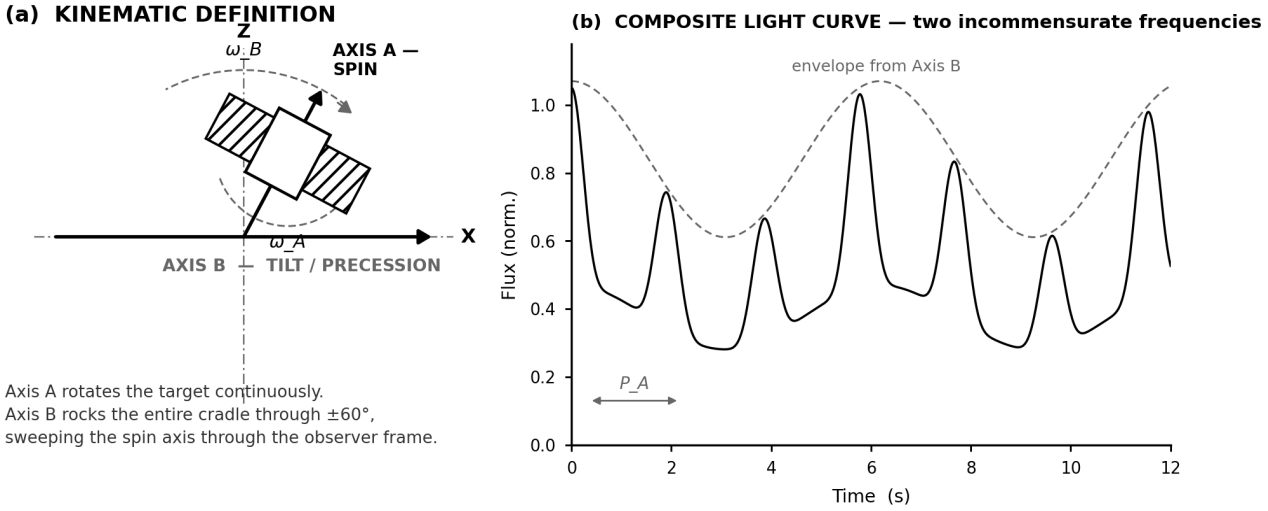


FIG. 3 — Kinematic definition and the resulting composite signal. Axis A spins the target; Axis B sweeps the spin axis. The two rates are chosen to be incommensurate, so the light curve does not repeat over the recording. The envelope carries the Axis B period; the fine structure carries the Axis A period.

5.1 Why a rocking cradle rather than a full gimbal

A true two-axis gimbal, in which the second axis rotates continuously rather than rocking, requires electrical power and signal to cross a continuously rotating joint. That means a **slip ring**. Slip rings introduce friction, contact noise on the sensor line, an additional failure mode, and a further procurement item.

The rocking cradle avoids this entirely. Because Axis B never exceeds ±60°, the cables serving Motor A and the slot sensor can pass through a fixed service loop that flexes rather than wraps. The kinematic benefit is preserved — two independent frequencies — while a component and a failure mode are removed. Given the schedule, this is the correct trade.

5.2 Rate selection

The two rates must be incommensurate, and must both be resolvable within the recording length and the frame rate. The following defaults are recommended for the first validation run:

Parameter	Value	Rationale
Axis A period P_A	1.85 s (≈ 32.4 rpm)	About 111 frames per revolution at 60 fps — ample to resolve the glint profile shape, not merely its timing.
Axis B period P_B	11.4 s	Ratio to P_A is 6.16 — deliberately non-integer, so the pattern does not repeat.
Recording length	120 s minimum	About 10 full Axis B cycles, giving a well-defined envelope frequency and useful error bars.
Frame rate	60 fps, constant	Above the Nyquist requirement for both frequencies with large margin.

AVOID INTEGER RATIOS
If P_B is set to an exact integer multiple of P_A, the two frequencies lock and the composite light curve becomes exactly periodic at the longer period. The inversion then appears to work perfectly while actually solving a much easier problem than intended. Always confirm the commanded ratio is irrational-looking to at least three decimal places before recording a validation run.

5.3 Motor rate budget

With 1.8° steppers at 1/16 microstepping, one output revolution requires 3,200 microsteps.

Axis	Output rate	Motor rate	Step rate	Margin
A (direct drive)	0.541 rev/s	0.541 rev/s	1,731 steps/s	Comfortable
B (3:1 belt reduction)	0.0877 rev/s	0.263 rev/s	842 steps/s	Comfortable
Combined load on Uno	—	—	2,573 steps/s	Below the ≈4,000 steps/s practical ceiling of AccelStepper on an ATmega328P

If a faster spin is wanted for a specific test, drop Axis A to 1/8 microstepping rather than raising the step rate — the ATmega328P is the binding constraint, not the motor.

SECTION 6

Mechanical design

The structure is 2020 aluminium T-slot extrusion on a rectangular base, carrying two vertical posts that support the Axis B shaft in rolling bearings. A cradle clamped to that shaft carries Motor A and the target. Motor B drives the shaft through a 3:1 GT2 belt reduction.

FIG. 4 FRONT ELEVATION — dimensions in mm

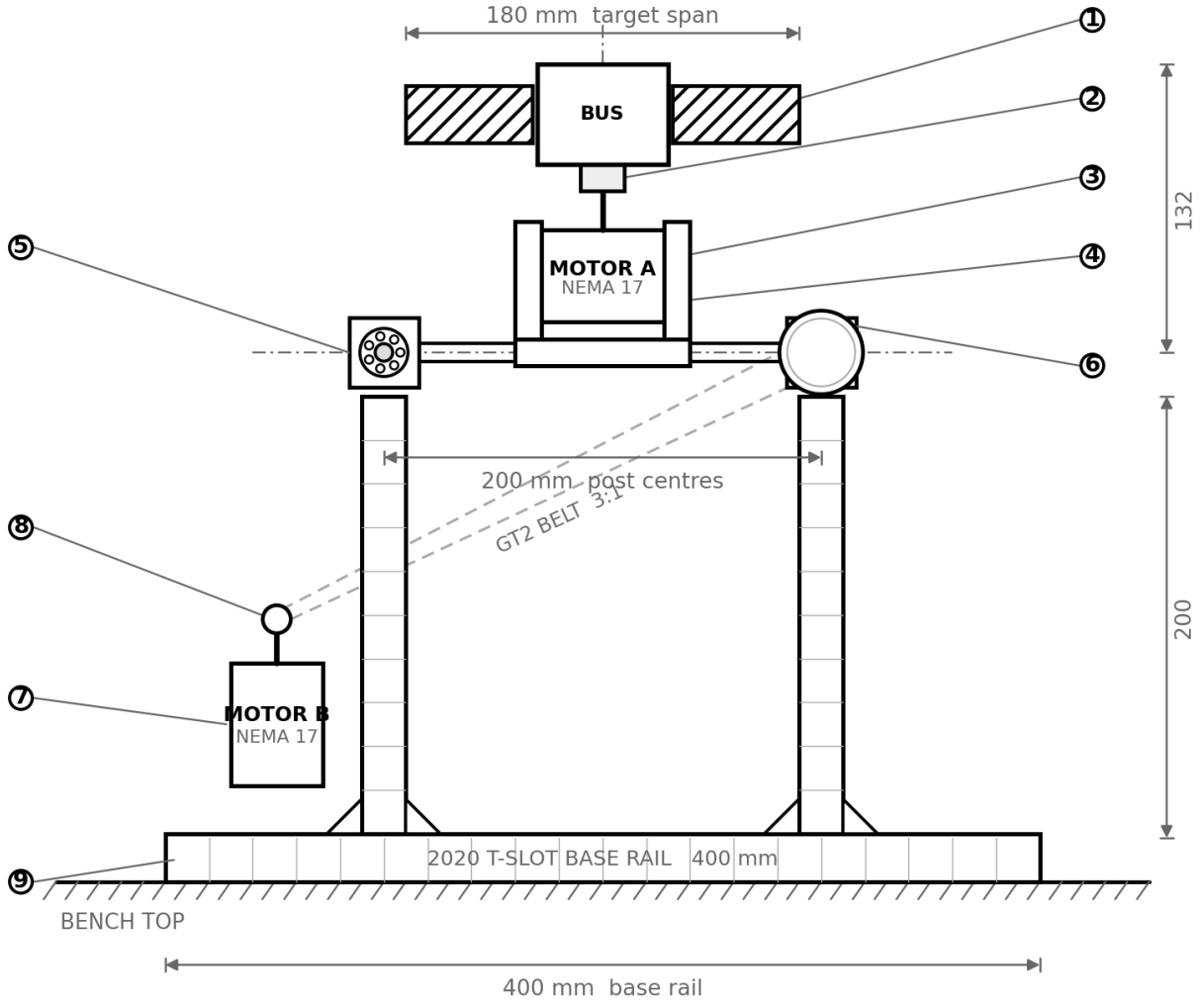


FIG. 4 — Front elevation. Item numbers refer to the balloon list in Table 6.1. The Axis B shaft is supported in 608ZZ bearings at both posts; the belt drive is routed behind the frame and is shown schematically.

6.1 Balloon item list

Item	Description	Qty	BOM code
1	Target model — printed bus with specular panels	1	G01–G05
2	Aluminium hub, 5 mm bore, M3 grub screw	1	B09
3	Motor A — NEMA 17, direct drive to target	1	A01
4	Cradle — U-bracket clamped to Axis B shaft	1	C11
5	Bearing block — 608ZZ in housing, both posts	2	B06, C11
6	Driven pulley — GT2 60T, 8 mm bore	1	B02
7	Motor B — NEMA 17, drives Axis B	1	A01
8	Drive pulley — GT2 20T, 5 mm bore	1	B01

Item	Description	Qty	BOM code
9	Base rail — 2020 extrusion, 400 mm	2	C01

FIG. 5 SIDE ELEVATION — Axis B sweep envelope, dimensions in mm

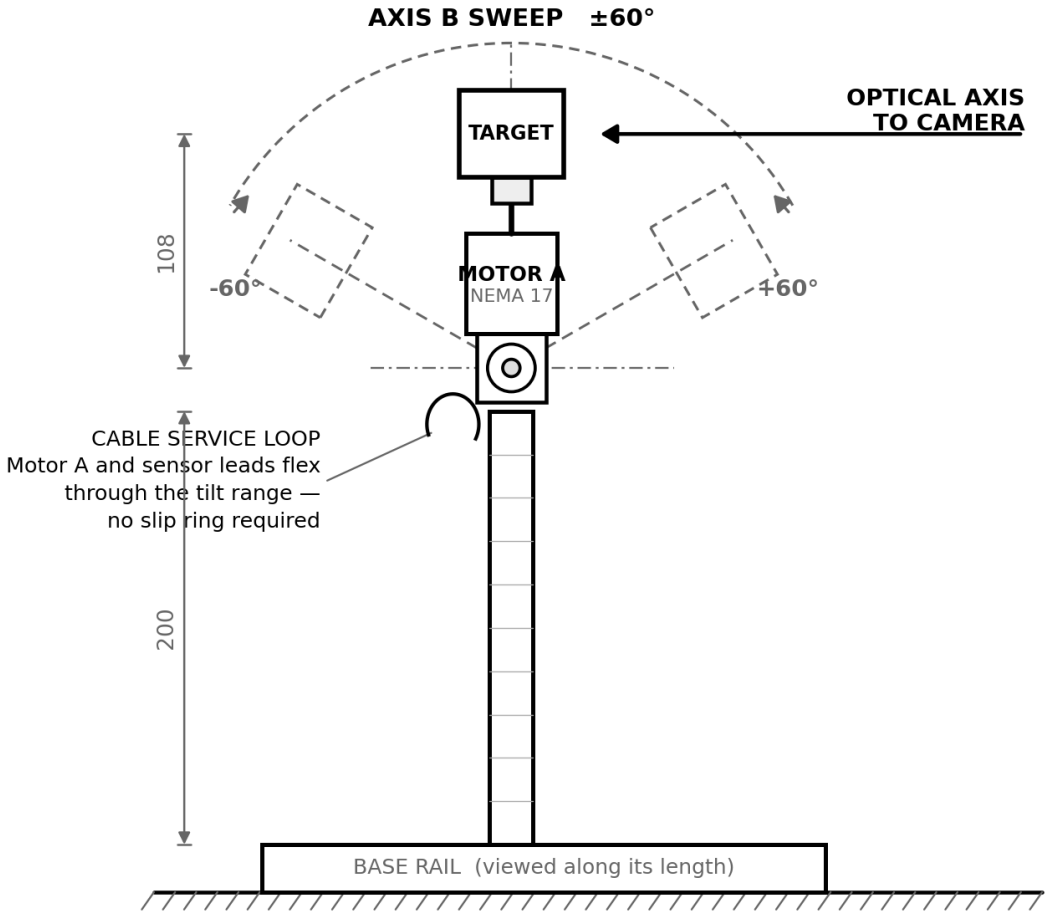


FIG. 5 — Side elevation showing the Axis B sweep envelope. The ±60° limit is what permits a flexing cable service loop in place of a slip ring. Ghost outlines show the target at the sweep extremes.

6.2 Critical mechanical requirements

Requirement	Value	Why it matters
Cradle balance about Axis B	Within ±5 g·cm	An unbalanced cradle loads Motor B asymmetrically, producing a rate error that is periodic at the Axis B frequency — precisely the frequency being measured. Counterweight until the cradle holds any position with power removed.
Target balance about Axis A	Visually balanced on the hub	Imbalance at the spin frequency induces vibration that blurs the glint profile.
Belt tension	Firm, deflects ≈2 mm under light thumb pressure at mid-span	Too loose gives tooth skip and a silent, undetected rate error. Too tight loads the motor bearing and raises audible noise.
Frame rigidity	No visible motion at the target under a light push	Frame flex couples motor vibration into the optical path.
Shaft end float	Constrained by two collars	Axial wander shifts the target within the camera field between runs.

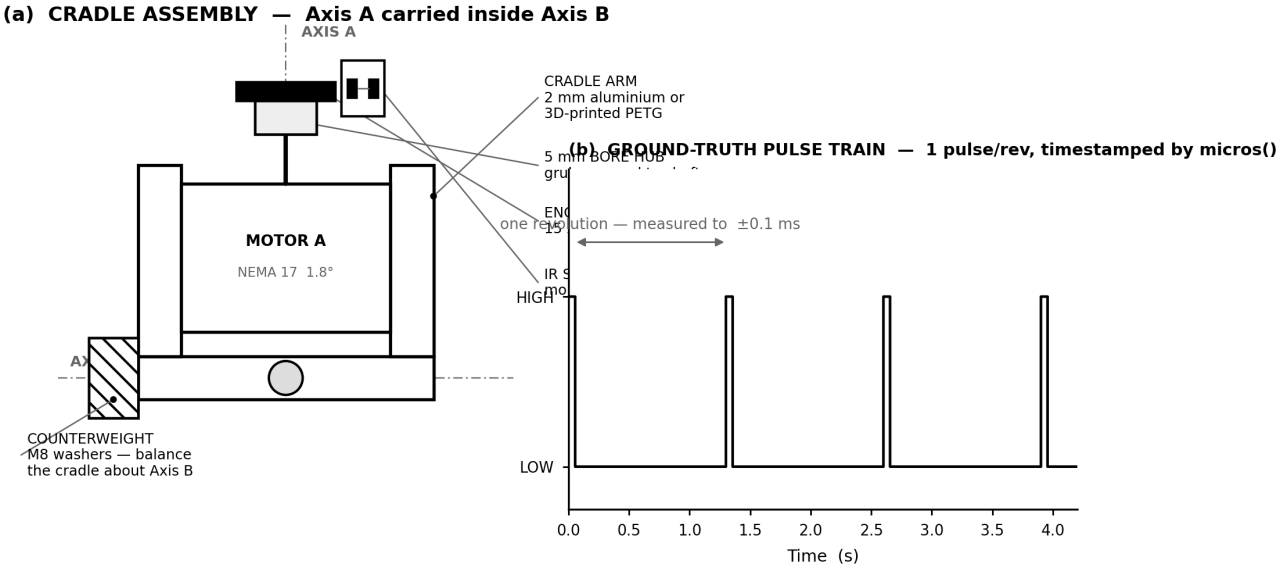


FIG. 6 — (a) Cradle assembly detail, showing Motor A carried inside the Axis B structure, the counterweight, and the encoder tab passing through the slot sensor. (b) The resulting ground-truth pulse train, one pulse per revolution of Axis A.

6.3 The ground-truth channel

The IR slot sensor is the least expensive item in the entire build and one of the two or three most important. A tab fixed to the target hub breaks the sensor's beam once per revolution; the controller timestamps each edge with `micros()` and streams the values over USB.

This produces the true rotation period of Axis A, measured by a completely independent physical channel — no camera, no video codec, no photometry. When the inversion pipeline reports a period, it is scored against this number. Without this channel the project can only demonstrate that its software produces *an* answer, not that the answer is *correct*.

> **TIMING RESOLUTION**
> Polling the sensor pin inside the stepper loop gives an edge resolution of order 20–100 µs. Against a 1.85 s period that is a fractional error below 0.006 %, which is far finer than the precision the optical inversion will achieve. The ground truth is therefore not the limiting term in the comparison — which is exactly the property a reference channel must have.

SECTION 7

Electrical design

Control is an Arduino Uno carrying a CNC Shield V3, which provides four StepStick driver sockets, screw terminals for motor power, and headers that expose the STEP and DIR lines. Two TMC2209 drivers occupy the X and Y sockets.

FIG. 7 ELECTRICAL INTERCONNECTION SCHEMATIC

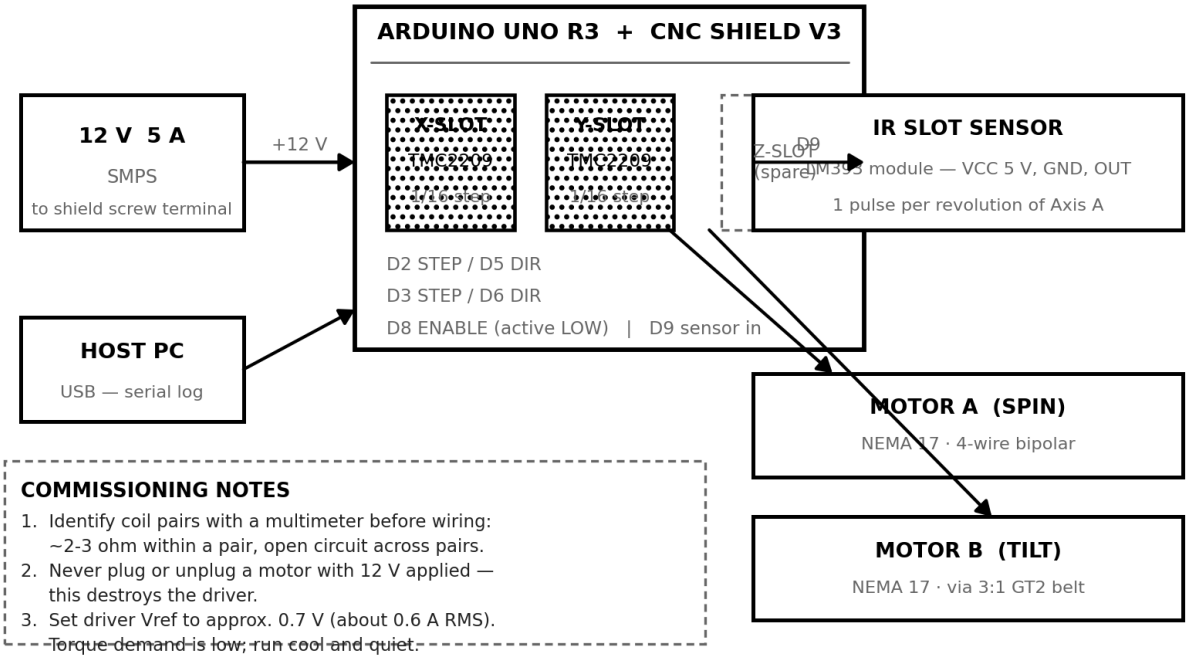


FIG. 7 — Electrical interconnection. The 12 V supply feeds the shield's motor rail; the Arduino is powered and logged over USB. The slot sensor returns to D9.

7.1 Why TMC2209 and not A4988

This is not a comfort upgrade and the extra cost is not optional. An A4988 at coarse microstepping produces measurable cogging — a periodic variation in instantaneous rotation rate, locked to the step frequency.

A periodic rate ripple is indistinguishable from a real periodic signal. It would appear in the periodogram as a spurious peak, and the team would spend days chasing a phantom frequency generated by its own motor. The TMC2209 in StealthChop mode with 1/16 microstepping and internal 256-step interpolation produces motion smooth enough that this artefact stays well below the measurement floor.

7.2 Driver configuration

Setting	Value	Method
Microstepping	1/16	MS1 and MS2 both pulled high — install jumpers in the first two positions under each driver socket. Vendor silkscreens differ; verify against the specific board's documentation before powering on.
Current (V_{REF})	Start at 0.70 V	Measure between the trimpot wiper and ground with the driver seated and 12 V applied but motors idle. Torque demand here is very low; run cool rather than strong.
Mode	StealthChop (default)	No configuration required for STEP/DIR operation.
Enable	D8, active LOW	Handled by the shield. Drive high to release both motors for manual positioning during setup.

THREE RULES THAT PREVENT THE MOST COMMON DESTRUCTIVE FAILURES

1. Never insert or remove a motor connector with 12 V applied. Back-EMF from the disconnecting coil destroys the driver. Power down first, every time.

2. Confirm driver orientation before applying power. A StepStick inserted backwards is destroyed instantly, and can take the shield with it.

3. Identify coil pairs with a multimeter before wiring — roughly 2–3 Ω within a pair, open circuit between pairs. Colour codes on NEMA 17 motors are not standardised across vendors, and a mis-paired motor will buzz and heat without turning.

7.3 Power budget

Load	Current	Note
Motor A at 0.6 A RMS per phase	≈0.85 A from 12 V	Two phases energised, chopper duty well below unity
Motor B at 0.6 A RMS per phase	≈0.85 A from 12 V	Same
COB LED, 10 W	≈0.83 A from 12 V	Continuous during recording
Arduino and sensor	≈0.10 A from USB	Not drawn from the 12 V rail
Total on the 12 V rail	**≈2.5 A**	A 5 A supply gives 100 % headroom. Do not substitute a smaller supply; rail sag under motor transients causes driver resets mid-recording.

7.4 Firmware behaviour

The controller sketch is deliberately minimal. It must do only four things, and adding anything else to the main loop risks disturbing step timing:

1. Accept two rate values over serial, in microsteps per second, one per axis.
2. Drive Motor A continuously at the commanded rate using `AccelStepper::runSpeed()`.
3. Drive Motor B in a triangular sweep between the ±60° limits, reversing at each limit, so that Axis B rocks at the commanded period.
4. Poll the slot-sensor pin every loop iteration; on a rising edge, print the `micros()` value over serial.

The host records the serial stream to a text file for the entire duration of the video recording. Cross-correlating the two afterwards is straightforward because both begin from a single operator action.

SECTION 8

Optical design

The optical chain is short, but three details in it decide whether the recorded data is usable at all. Each is cheap to get right and expensive to discover late.

FIG. 8 OPTICAL LAYOUT — PLAN VIEW (schematic; distances annotated, not to scale)

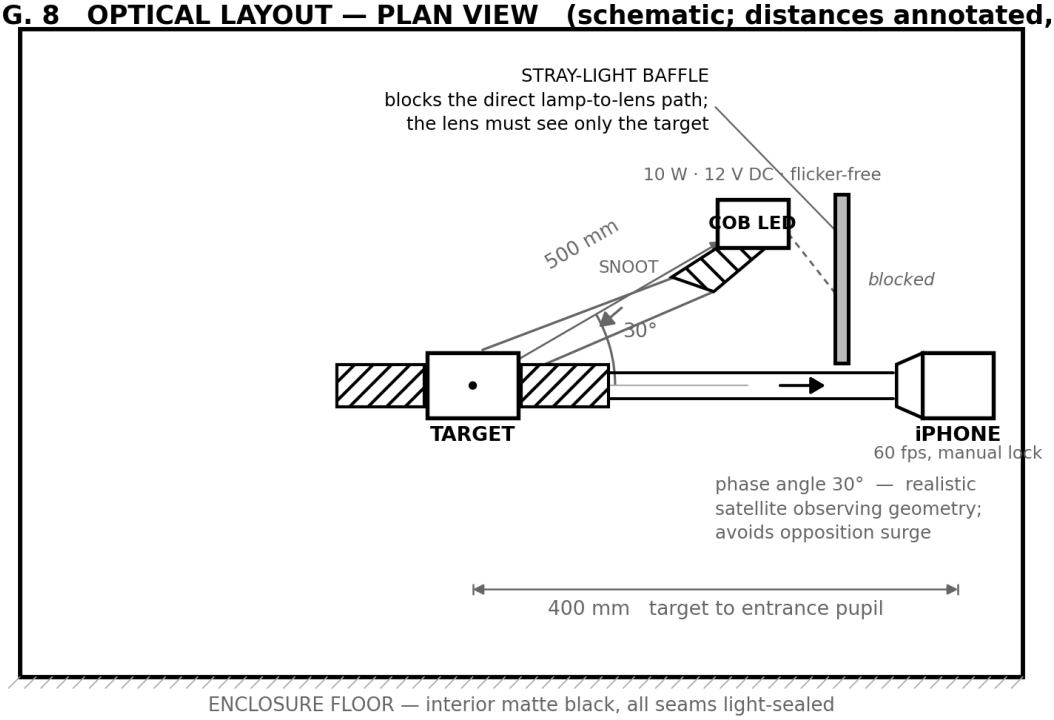


FIG. 8 — Optical layout in plan view. The lamp illuminates the target at a 30° phase angle; a baffle prevents any direct lamp-to-lens path. All interior surfaces are matte black.

8.1 Illumination — the flicker trap

A mains-powered LED floodlight must not be used. Inexpensive LED floods run from rectified mains and modulate at 100 Hz. Recording at 60 fps produces a beat between the sampling rate and the lamp modulation, which appears in the light curve as a strong periodic signal that is entirely an artefact of the lighting.

This failure is insidious because the artefact *looks like a real detection*. A team could reasonably report a spurious period, defend it confidently, and be wrong. A 10 W COB LED module driven from the same regulated 12 V DC supply as the motors emits constant flux and removes the problem for about ₹250.

Illumination requirement	Specification
Supply	12 V DC from the regulated SMPS. Never mains-driven, never PWM-dimmed.
Colour temperature	≈5000 K, so the specular and diffuse components are captured with comparable sensor response across channels.
Angular extent	Restricted by a snoot to approximately the target envelope, so the enclosure walls are not illuminated.
Thermal	COB bonded to a 40 mm heatsink with thermal adhesive. An unheatsunk 10 W COB will droop in output over a few minutes, introducing a slow downward trend into every light curve.
Phase angle	30° between the illumination direction and the viewing direction, measured at the target.

8.2 Why 30° and not 0°

Placing the lamp beside the camera — near-zero phase angle — is tempting because it maximises returned flux. It is wrong for two reasons. First, the camera physically shadows the target. Second, at near-zero phase most surfaces exhibit an opposition brightening that compresses the contrast between specular and diffuse components, which is the very contrast being measured. A 30° geometry is both realistic for satellite observation and better conditioned for the inversion.

8.3 Stray light

Any light reaching the sensor that did not reflect off the target adds a constant offset to every frame. That offset does not merely raise the baseline — it suppresses the fractional modulation depth, which is the quantity carrying the geometric information. A target modulating between 0.1 and 1.0 in true flux, sitting on a stray-light pedestal equal to its own peak, modulates only between 1.1 and 2.0 as recorded: the amplitude ratio collapses from 10:1 to under 2:1.

— Interior surfaces painted matte black or lined with black chart paper.

— A baffle positioned to block every direct path from lamp to lens, verified by eye from the lens position with the target removed.

— All enclosure seams taped. Record with room lights off.

— **Verification test:** record 10 s with the target removed and the lamp on. The mean frame level of that recording is the stray-light floor. It must be below 2 % of the target's peak level. If not, find the leak before recording anything else.

8.4 Camera configuration

The iPhone is an entirely adequate photometric sensor for this task, provided every automatic behaviour is defeated. Automatic exposure is the single largest threat to the measurement: it responds to exactly the brightness variations being recorded, and actively flattens them.

Setting	Required state	How
Capture app	Blackmagic Camera (free)	The stock Camera app does not expose locked manual ISO and shutter for video.
ISO	Locked, manual	Set once during setup; never touched between runs.
Shutter	Locked, manual	Choose the longest exposure that does not clip the glint peak.
White balance	Locked	Auto white balance drifts with scene content.
Focus	Locked	Autofocus hunting changes the point-spread function mid-recording.
Smart HDR	OFF — Settings → Camera	Applies per-frame non-linear tone mapping. Photometrically fatal.
Frame rate	60 fps, constant	Preferred over 240 fps slow motion, which surrenders exposure control on most devices.
Codec	ProRes if available, else HEVC at maximum bitrate	The target is bright against black — the easy case for a codec, but headroom helps.

AN HONEST LIMITATION TO STATE IN THE WRITE-UP

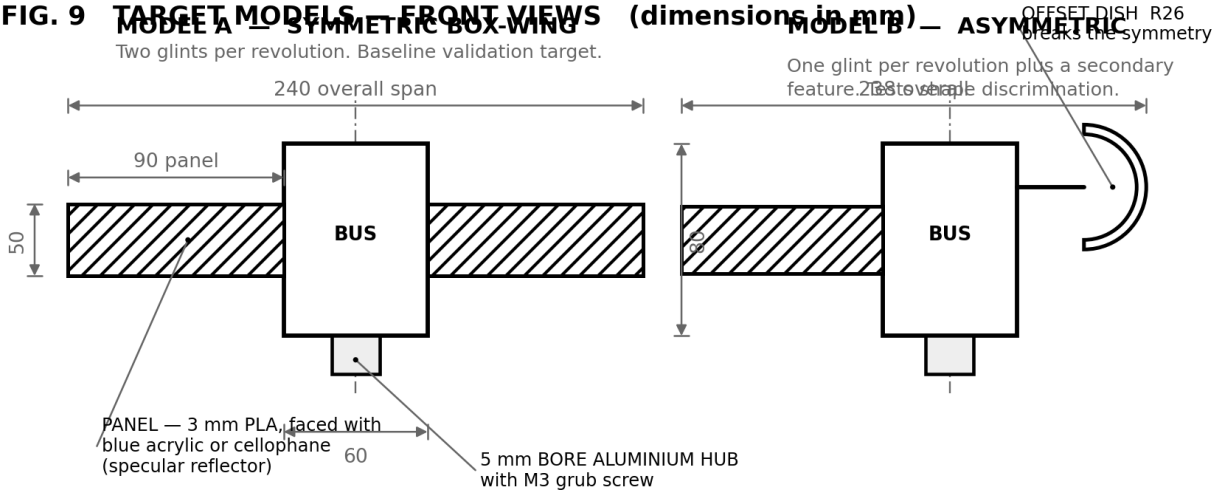
iPhone video is gamma-encoded and tone-mapped. Recorded pixel values are therefore **not linear** in incident flux. Period determination is unaffected, because periodicity survives any monotonic transform. Shape and area inversion *is* affected. The mitigation is the printed grayscale step wedge fixed inside the field of view: photographing a known reflectance ladder in every frame permits an empirical response curve to be fitted and inverted. Stating this limitation and its mitigation is stronger than concealing it — an examiner who spots an unacknowledged non-linearity will discount the whole result.

SECTION 9

Target models

Two targets are printed. The first is symmetric and serves as the baseline validation article; the second breaks that symmetry deliberately, to test whether the inversion can discriminate between shapes rather than merely recover a period.

FIG. 9 TARGET MODELS — FRONT VIEWS (dimensions in mm)



PRINT SETTINGS
PLA · 0.2 mm layer · 15 % gyroid infill · 2 perimeters · no supports if the bus is printed open-face down. Bus approx. 8 h, each panel approx. 1 h. Total mass approx. 75 g — well within the NEMA 17 radial load. Balance the assembled model on the hub before running Axis A above 0.5 rev/s.

FIG. 9 — Target model front views with print settings. Model A is the symmetric box-wing baseline; Model B carries a single panel and an offset dish that breaks rotational symmetry.

9.1 Why printing does not remove the surfacing work

Bare PLA is close to a uniform Lambertian diffuser in every direction. A target printed and left untreated produces a low-contrast, nearly featureless light curve with very little structure to invert. The entire measurement depends on the **contrast** between a strongly specular panel and a diffusely reflecting bus.

— **Bus:** wrap in aluminised mylar tape, deliberately creased. Creasing produces the irregular multi-lobed scattering characteristic of real multilayer insulation, rather than a single mirror facet.

— **Panels:** face with blue-tinted acrylic offcut or blue cellophane over the printed substrate. This gives the flat, strongly directional specular return that generates the glints.

— **Hub:** use a metal hub with a grub screw, not a printed bore. Printed bores strip under repeated mounting and the resulting eccentricity is a recurring, hard-to-diagnose source of vibration.

9.2 Two models, and why the second one earns its print time

Recovering the parameters of one model demonstrates that the pipeline works on the case it was tuned for. Demonstrating that the pipeline distinguishes Model A from Model B — that the fitted geometry differs in the right direction, by the right sense — is a substantially stronger claim, and it is the difference between a working demo and a result.

	Model A	Model B
Geometry	Symmetric box-wing, two opposed panels	Single panel plus offset dish
Glints per revolution	Two	One dominant, plus a weaker secondary feature
Harmonic signature	Strong power at $2/P_A$, weak at $1/P_A$	Strong power at $1/P_A$

	Model A	Model B
Purpose	Baseline validation; harmonic-ambiguity test case	Shape discrimination; tests that the fit is sensitive to asymmetry
Print time	≈8 h bus + 2 h panels	≈8 h bus + 1 h panel + 2 h dish

START THE PRINTS ON DAY 1

The bus is an eight-hour print and printers fail overnight without warning. Beginning both prints on day one costs nothing — the machine runs unattended while procurement and software work proceed — and leaves time to reprint a failure. A print started on day 4 that fails on day 5 removes a day from a schedule that has only one spare.

SECTION 10

Software

The analysis chain has nine stages. Stages 5 through 8 — the actual science — can and should be written and validated against **synthetic data** before any hardware exists.

FIG. 10 SOFTWARE PIPELINE

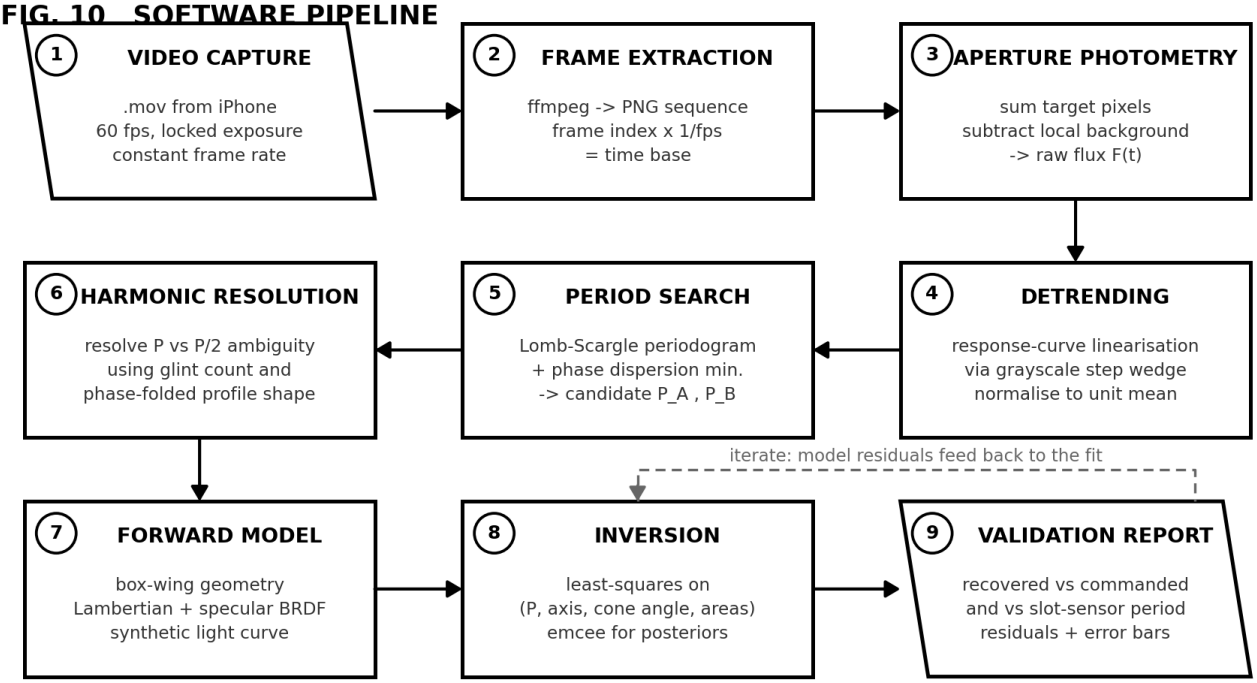


FIG. 10 — Software pipeline. Stages 5 to 8 are hardware-independent and are developed first against a simulated light curve, so that parts delays cannot stall the analysis work.

10.1 The synthetic-data strategy

This is the single most schedule-protective decision in the project. A forward model that generates a light curve from known parameters is perhaps forty lines of Python: a box-wing geometry, a Lambertian plus specular reflection term, a commanded spin and precession, sampled at 60 Hz with noise added.

Feeding that synthetic curve into the period search and the inversion validates the entire analysis chain on day one, using known inputs, with no hardware. When real data eventually arrives, only the data source changes. Parts delays stop being fatal, and the hardest debugging — getting the fit to converge — happens while there is still schedule to absorb it.

10.2 Stage detail

Stage	Tool	Detail
1 Capture	Blackmagic Camera	Locked exposure, 60 fps CFR, ≥120 s.
2 Frame extraction	ffmpeg	`ffmpeg -i in.mov -vsync 0 f%06d.png`. Verify the extracted frame count equals duration × 60 exactly; a mismatch indicates variable frame rate and invalidates the time base.
3 Photometry	OpenCV, photutils	Fixed circular aperture enclosing the full target envelope at all attitudes, plus an annular background region. Sum minus scaled background gives raw flux.
4 Detrending	NumPy	Fit the camera response curve from the grayscale step wedge; invert it. Normalise to unit mean. Remove any residual linear drift.
5 Period search	astropy LombScargle	Periodogram over 0.02–2.5 Hz, plus phase dispersion minimisation as an independent check. Expect peaks at $1/P_A$, $2/P_A$ and $1/P_B$.
6 Harmonic resolution	Custom	Fold at candidate P and at 2P; the correct period produces a single self-consistent profile, the incorrect one produces two alternating profiles. This is where Model B is diagnostic.

Stage	Tool	Detail
7 Forward model	NumPy	Box-wing with Lambertian plus specular terms; parameters are P_A, P_B, cone angle, phase offsets, panel area and diffuse albedo.
8 Inversion	scipy.optimize, emcee	Least squares for the point estimate, then MCMC for posteriors. Report medians and 16th/84th percentiles, not bare best-fit values.
9 Validation	pandas, matplotlib	Recovered against commanded and against slot-sensor period. Residual plot. Error table.

REPORT UNCERTAINTIES, NOT POINT ESTIMATES

A recovered period quoted as 1.847 s is an assertion. A recovered period quoted as 1.847 ± 0.004 s, against a commanded 1.850 s and a slot-sensor measurement of 1.8502 s, is a measurement. The difference costs one MCMC run and is what distinguishes this project from a demonstration.

SECTION 11

Bill of materials

Every item required to build the instrument is listed below. Quantities include the spares identified in Section 16 as schedule-protective. Prices are indicative retail figures for the Delhi NCR market and Indian online retailers at the time of issue; treat them as budgeting guidance, not quotations.

RECONCILIATION WITH EARLIER ESTIMATES
Earlier working estimates for this build were approximately ₹6,300 and then ₹10,460. Both were low, and for the same reason: they costed the **mechanism** and omitted the **assembly** — couplers, brackets, fasteners, connectors, heatsinks, adhesives and consumables. That category routinely accounts for 30–40 % of a real build and is the most common source of BOM underestimation. The figures below are the exhaustive count and supersede both earlier numbers.

11.1 Itemised parts list

Code	Item	Specification	Qty	Unit ₹	Line ₹	Function in the build	Source
GROUP A — MOTION AND CONTROL							
A01	Stepper motor, NEMA 17	17HS4401 · 1.8° · 40 N·cm · 1.5 A/phase · 4-wire bipolar · 5 mm shaft	2	650	1,300	Axis A spin drive; Axis B tilt drive	Market
A02	Stepper driver, TMC2209	StepStick footprint · UART/legacy STEP-DIR · with heatsink	2	300	600	Smooth microstepped motion; prevents cogging artefacts in the light curve	Market / Amazon
A03	Stepper driver, TMC2209 — SPARE	As A02	2	300	600	Schedule protection; a failed driver mid-build costs a day	Market / Amazon
A04	Arduino Uno R3	ATmega328P · USB-B	1	450	450	Motion control and slot-sensor timestamping	Market
A05	Arduino Uno R3 — SPARE	As A04	1	400	400	Single point of failure for the entire motion system	Market
A06	CNC Shield V3	4 × StepStick sockets · screw terminal · microstep jumpers	1	300	300	Driver carrier; exposes STEP/DIR and limit headers	Amazon / Robu
A07	Driver heatsink, adhesive	9 × 9 × 5 mm aluminium	4	15	60	Thermal management for TMC2209 packages	Market
A08	SMPS power supply	12 V · 5 A · 60 W · centre-positive barrel output	1	550	550	Single regulated rail for both motors and the COB LED	Market
A09	DC barrel jack, female, screw-terminal	5.5 × 2.1 mm	2	25	50	Supply distribution to shield and LED	Market
A10	USB cable, A to B	1.5 m	1	120	120	Host link — power, upload, serial log	Market
Group A subtotal					**4,430**		
GROUP B — TRANSMISSION AND ROTATING HARDWARE							
B01	GT2 timing pulley, 20 T	5 mm bore · 6 mm belt width · with grub screws	1	120	120	Motor B drive pulley	Amazon / Robu
B02	GT2 timing pulley, 60 T	8 mm bore · 6 mm belt width	1	280	280	Axis B driven pulley — gives the 3:1 reduction	Amazon / Robu
B03	GT2 belt, closed loop	280 mm (140 T) · 6 mm wide	1	150	150	Motor B to Axis B shaft	Amazon / Robu
B04	GT2 belt — SPARE	300 mm (150 T) · 6 mm wide	1	150	150	Allows centre-distance adjustment if the frame is built off-nominal	Amazon / Robu
B05	Flexible shaft coupler	5 mm to 8 mm · aluminium, helical cut	1	150	150	Accommodates minor misalignment without binding	Amazon / Robu
B06	Ball bearing, 608ZZ	8 mm ID · 22 mm OD · 7 mm width · shielded	4	40	160	Axis B shaft support at both posts; two spare	Market

Code	Item	Specification	Qty	Unit ₹	Line ₹	Function in the build	Source
B07	Steel shaft, ground	8 mm dia. × 200 mm · h6 tolerance	1	120	120	Axis B rotation shaft	Market
B08	Shaft collar with grub screw	8 mm bore	2	60	120	Constrains shaft end float	Market
B09	Aluminium hub, grub-screw	5 mm bore · M3 grub · flange for target mounting	1	180	180	Mounts the printed target to the Motor A shaft	Amazon / Robu
Group B subtotal					**1,430**		
GROUP C — STRUCTURE AND FASTENERS							
C01	Aluminium extrusion, 2020	400 mm length · T-slot	2	110	220	Base rails, long sides	Market
C02	Aluminium extrusion, 2020	210 mm length · T-slot	2	60	120	Base rails, cross members	Market
C03	Aluminium extrusion, 2020	200 mm length · T-slot	2	58	116	Vertical posts carrying the Axis B bearings	Market
C04	Corner bracket, 2020 L-type	With fixing holes, black	12	22	264	Frame joints — base corners and post feet	Market
C05	T-nut, M5, for 2020 slot	Drop-in or slide-in	40	4	160	All extrusion fixings	Market
C06	Bolt, M5 × 10, socket cap	Stainless or zinc	40	3	120	Mates with C05	Market
C07	Bolt, M3 × 10, socket cap	For NEMA 17 face mounting	20	2	40	Motor and bracket fixings	Market
C08	Nut, M3	Plain	20	1	20	General fixing	Market
C09	Washer, M3	Plain	20	1	20	Load spreading on printed parts	Market
C10	NEMA 17 mounting bracket, L-type	Steel, slotted for belt tensioning	2	120	240	Motor A in cradle; Motor B on base with tension adjustment	Amazon / Robu
C11	Cradle and bearing housings	3D printed PETG, or 2 mm aluminium sheet	1	0	0	Carries Motor A; retains 608ZZ bearings in the posts	Printed
C12	Washer, M8, heavy	For counterweight stack	20	3	60	Balances the cradle about Axis B	Market
C13	Stud M8 × 40 with two nuts	Threaded rod section	1	40	40	Counterweight mounting	Market
C14	Rubber foot, self-adhesive	20 mm dia.	4	10	40	Vibration isolation from the bench	Market
Group C subtotal					**1,460**		
GROUP D — GROUND-TRUTH SENSING							
D01	IR slot / opto-interrupter module	LM393 comparator · 5 V · digital OUT · 10 mm slot	1	80	80	One pulse per revolution of Axis A — the independent truth channel	Market / Robu
D02	IR slot sensor — SPARE	As D01	1	80	80	Critical measurement path; spare is mandatory	Market / Robu
D03	Encoder tab	15 × 10 × 1 mm, opaque	1	0	0	Breaks the sensor beam once per revolution	Printed
Group D subtotal					**160**		
GROUP E — ILLUMINATION AND ENCLOSURE							
E01	COB LED module	10 W · 12 V DC direct-drive · ≈5000 K	1	250	250	Flicker-free solar simulator — the single most important optical choice	Market
E02	COB LED module — SPARE	As E01	1	250	250	Thermal failure of a COB is silent and mid-run	Market
E03	Aluminium heatsink	40 × 40 × 11 mm, finned	1	90	90	Prevents output droop over a recording; unheatsunk COBs drift downward	Market
E04	Thermal adhesive pad or paste	Small tube / 20 × 20 mm pad	1	60	60	Bonds E01 to E03	Market

Code	Item	Specification	Qty	Unit ₹	Line ₹	Function in the build	Source
E05	Blackout cloth	Cotton, matte black, 2 m × 1.5 m	1	380	380	Enclosure lining and light sealing	Market
E06	Black chart paper	A2, matte	10	15	150	Interior surfacing, snoot, baffle facing	Stationers
E07	Matte black spray paint	200 ml, acrylic, ultra-matte	1	240	240	Interior surfaces and any reflective hardware in the field of view	Market
E08	Foam board	5 mm, A3	2	60	120	Stray-light baffle and snoot structure	Stationers
E09	Grayscale step wedge	Printed reflectance ladder on matte paper	1	20	20	Camera response linearisation reference, kept in-frame	Print shop
E10	Cardboard carton	≈600 × 450 × 450 mm	1	0	0	Enclosure shell	Salvage
Group E subtotal					**1,560**		
GROUP F — CAMERA SUPPORT							
F01	Tripod	≈1.3 m · pan-tilt head · 1/4-20 thread	1	700	700	Fixed, repeatable camera geometry between runs	Amazon / Market
F02	Phone clamp	Universal, 1/4-20 socket	1	200	200	Mounts the iPhone to F01	Amazon / Market
F03	Spirit level, small	Bubble, adhesive or shoe-mount	1	60	60	Repeatable camera attitude	Market
Group F subtotal					**960**		
GROUP G — TARGET MODELS							
G01	PLA filament	1.75 mm · white · 1 kg spool	1	900	900	Both target buses and all panels	Amazon / Robu
G02	Aluminised mylar tape	48 mm × 10 m	1	180	180	Multilayer-insulation surrogate on the bus — diffuse, multi-lobed return	Market
G03	Blue transparent facing	Blue acrylic offcut 100 × 200 × 2 mm, or blue cellophane	1	150	150	Solar-panel surrogate — provides the specular glint component	Market
G04	Grub screw, M3 × 4	Cup point	4	5	20	Hub retention	Market
G05	Double-sided foam tape	12 mm × 5 m	1	70	70	Panel and facing attachment	Market
Group G subtotal					**1,320**		
GROUP H — WIRING AND INTERCONNECT							
H01	Hookup wire, stranded	22 AWG · 6 colours · 5 m each	1	280	280	Motor extensions, sensor leads, LED supply	Market
H02	Jumper wires, male-male	40-way ribbon, 20 cm	1	70	70	Shield-to-header connections	Market
H03	Jumper wires, male-female	40-way ribbon, 20 cm	1	70	70	Sensor module to shield header	Market
H04	Jumper wires, female-female	40-way ribbon, 20 cm	1	70	70	General interconnect	Market
H05	JST-XH connector pairs, 4-pin	With crimp terminals	4	15	60	Motor leads — permits safe disconnection with power off	Market
H06	Screw terminal block, 2-pin	5 mm pitch, PCB or panel	4	12	48	12 V distribution to LED and shield	Market
H07	Heat-shrink tubing	Assorted 2–6 mm	1	90	90	All solder joints	Market
H08	Spiral cable wrap	6 mm × 2 m	1	60	60	Forms and protects the cradle service loop	Market
H09	Cable tie, 100 mm	Nylon	50	0.8	40	Cable management and strain relief	Market
H10	Adhesive cable clip	Small	10	3	30	Routes the service loop away from moving parts	Market
Group H subtotal					**818**		

Code	Item	Specification	Qty	Unit ₹	Line ₹	Function in the build	Source
GROUP I — CONSUMABLES							
I01	Hot glue sticks	11 mm	10	6	60	Target assembly and non-structural fixings	Market
I02	Solder, 60/40 rosin core	0.8 mm · 50 g	1	150	150	All permanent joints	Market
I03	Electrical tape	PVC, black	2	20	40	Insulation and light sealing	Market
I04	Masking tape	24 mm	1	40	40	Alignment marks, temporary fixing, seam sealing	Market
I05	Sandpaper	400 and 800 grit	2	15	30	Print finishing at mating surfaces	Market
I06	Isopropyl alcohol	99 % · 100 ml	1	90	90	Cleaning optical surfaces and print beds	Market
I07	Threadlocker, medium strength	Blue · 5 ml	1	120	120	All grub screws — vibration will loosen them otherwise	Market
I08	Lint-free cloth / cotton buds	Pack	1	50	50	Optical cleaning	Market
Group I subtotal					**580**		

11.2 Cost summary

Group	Category	Subtotal ₹	Share
A	MOTION AND CONTROL	4,430	34.8 %
B	TRANSMISSION AND ROTATING HARDWARE	1,430	11.2 %
C	STRUCTURE AND FASTENERS	1,460	11.5 %
D	GROUND-TRUTH SENSING	160	1.3 %
E	ILLUMINATION AND ENCLOSURE	1,560	12.3 %
F	CAMERA SUPPORT	960	7.5 %
G	TARGET MODELS	1,320	10.4 %
H	WIRING AND INTERCONNECT	818	6.4 %
I	CONSUMABLES	580	4.6 %
TOTAL	**FULL BUILD, INCLUDING ALL SPARES**	**12,718**	**100 %**

11.3 Cost-reduction variants

Variant	Change from full build	Total ₹
Full build	As specified above, all spares included	12,718
Filament already available	Remove G01 (−900)	11,818
Lean build	Remove all spares (−1,480); plywood baseboard instead of extrusion (−650); chart paper instead of spray paint (−240); budget tripod (−350); filament on hand (−900)	9,098

DO NOT CUT THE SPARES
The spare drivers, Arduino, belt, slot sensor and COB LED total ₹1,480 — about 12 % of the build. Against an eleven-day schedule with a single contingency day, one failed component without a replacement on the bench costs a market trip and most of a working day. The spares buy back more schedule than they cost in rupees, and schedule is the scarcest resource in this project.

SECTION 12

Assembly procedure

Build in this order. Each phase ends with a checkpoint that must pass before proceeding — the sequence is arranged so that a failure is detected while it is still cheap to correct.

12.1 Phase 1 — Structure

1. Cut or have cut: two 400 mm rails, two 210 mm cross members, two 200 mm posts from 2020 extrusion. Deburr all cut ends — burrs prevent brackets seating square.

2. Assemble the 400 × 250 mm base rectangle using eight corner brackets, M5 T-nuts and bolts. Do not fully tighten yet.

3. Check the base for square by comparing diagonals. Adjust, then tighten fully.

4. Mount the two posts vertically at 200 mm centres using the remaining four brackets. Check each for plumb against a spirit level in two planes.

5. Fit the four rubber feet.

> **CHECKPOINT 1**
> The frame stands without rocking, the posts are plumb, and pushing the top of a post by hand produces no visible deflection at the bearing height.

12.2 Phase 2 — Axis B

6. Press one 608ZZ bearing into each bearing housing (printed or fabricated). The fit should be firm; if loose, a wrap of tape on the outer race is acceptable.

7. Bolt the housings to the posts so the two bores are coaxial. Check by sliding the 8 mm shaft through both — it must pass without force.

8. Fit the 60 T pulley to the shaft near the right-hand post; secure the grub screw with threadlocker.

9. Fit a shaft collar outboard of each bearing to constrain end float. Leave approximately 0.2 mm axial play — a shaft clamped hard between collars will bind.

10. Mount Motor B to the base on its slotted L-bracket, fit the 20 T pulley, and install the 280 mm belt.

11. Tension the belt by sliding the motor bracket. Correct tension deflects about 2 mm under light thumb pressure at mid-span.

> **CHECKPOINT 2**
> With power off, rotating the 20 T pulley by hand turns the shaft smoothly through the full ±60° range with no binding, no belt skip, and no audible bearing roughness.

12.3 Phase 3 — Axis A and the cradle

11. Assemble the cradle and clamp it to the Axis B shaft at mid-span. Verify it does not foul either post across the full sweep.

12. Mount Motor A inside the cradle with its shaft perpendicular to the Axis B shaft.

13. Fit the aluminium hub to the Motor A shaft; threadlock the grub screw.

14. Fit the encoder tab to the hub and mount the slot sensor to the cradle so the tab passes cleanly through the slot with roughly 1 mm clearance on both sides.

15. Fit the counterweight stud and washers on the opposite side of the cradle.

16. Balance: with power off and the drive belt released, the cradle must hold any position through the sweep. Add or remove washers until it does.

17. Route Motor A and sensor cables through a spiral-wrapped service loop, clipped so nothing can catch at either sweep extreme. Sweep by hand through the full range and watch the cable.

CHECKPOINT 3
The cradle is balanced, the tab interrupts the sensor once per revolution, and the cable loop flexes freely at both ±60° limits without tension or snagging.

12.4 Phase 4 — Electronics

18. With everything unpowered, identify each motor's coil pairs with a multimeter: approximately 2–3 Ω within a pair, open circuit across pairs. Label the leads.

19. Fit both TMC2209 drivers to the X and Y sockets. **Confirm orientation against the shield silkscreen before applying power.**

20. Install microstepping jumpers for 1/16 on both sockets.

21. Connect motors through JST connectors. Connect the slot sensor to 5 V, GND and D9.

22. Apply 12 V with motors connected but the sketch not yet running. Set each driver's V_{REF} to approximately 0.70 V.

23. Upload the motion sketch. Command a slow rate on each axis independently and confirm direction and smoothness.

CHECKPOINT 4
Both axes run smoothly at commanded rates, drivers are warm but not hot to a brief touch, and the serial monitor shows one timestamp per revolution of Axis A with consistent intervals.

12.5 Phase 5 — Optical

24. Line the enclosure interior with black chart paper; paint any reflective hardware inside the field of view matte black.

25. Bond the COB LED to its heatsink with thermal adhesive. Wire to 12 V. Confirm no visible flicker by recording 5 s at 60 fps and inspecting the frame-mean trace — it must be flat.

26. Position the lamp at approximately 500 mm and 30° from the viewing axis; fit the snoot.

27. Position the camera at 400 mm on the tripod; level it and mark the tripod feet so the geometry is repeatable.

28. Install the baffle. From the lens position, with the target removed, confirm no direct view of the lamp.

29. Fix the grayscale step wedge inside the frame, outside the target's swept envelope.

30. Run the stray-light test: record 10 s with the target removed and the lamp on. Mean level must be under 2 % of the target's peak.

CHECKPOINT 5
Stray light is below 2 % of peak, the lamp shows no flicker in a 60 fps recording, and the target remains fully within frame across the entire ±60° sweep.

SECTION 13

Commissioning and calibration

13.1 Camera setup — perform once, then never touch

1. Open Blackmagic Camera. Set 1080p at 60 fps.
2. Frame the target with margin for the full sweep. Lock focus on the target at nominal attitude.
3. Command Axis A to run. Adjust ISO and shutter so the brightest glint peak sits at approximately 85 % of full scale — bright enough for signal-to-noise, below clipping.
4. Lock ISO, shutter and white balance. Confirm Smart HDR is off in iOS Settings → Camera.
5. Record 10 s, transfer, and inspect the frame-mean trace. It must show the expected modulation and no drift. Any slow downward trend indicates LED thermal droop — allow the lamp to reach thermal equilibrium for 10 minutes before recording data.

> **CLIPPING IS UNRECOVERABLE**
> If the glint peak saturates, the peak flux is lost and cannot be reconstructed by any post-processing. The specular amplitude is the parameter carrying most of the geometric information, so a clipped recording is worthless for shape inversion even though it still yields a period. Check the histogram at setup, and re-check after any change to the target or lamp position.

13.2 Photometric linearisation

6. Photograph the grayscale step wedge under the run illumination, in frame.
7. Measure the mean recorded level of each patch.
8. Fit recorded level against nominal patch reflectance — a power law is normally adequate.
9. Invert this response and apply it to all subsequent photometry before the inversion stage.
10. State the fitted exponent in the results. It is a property of the camera pipeline and reporting it is part of an honest measurement.

13.3 Validation run sequence

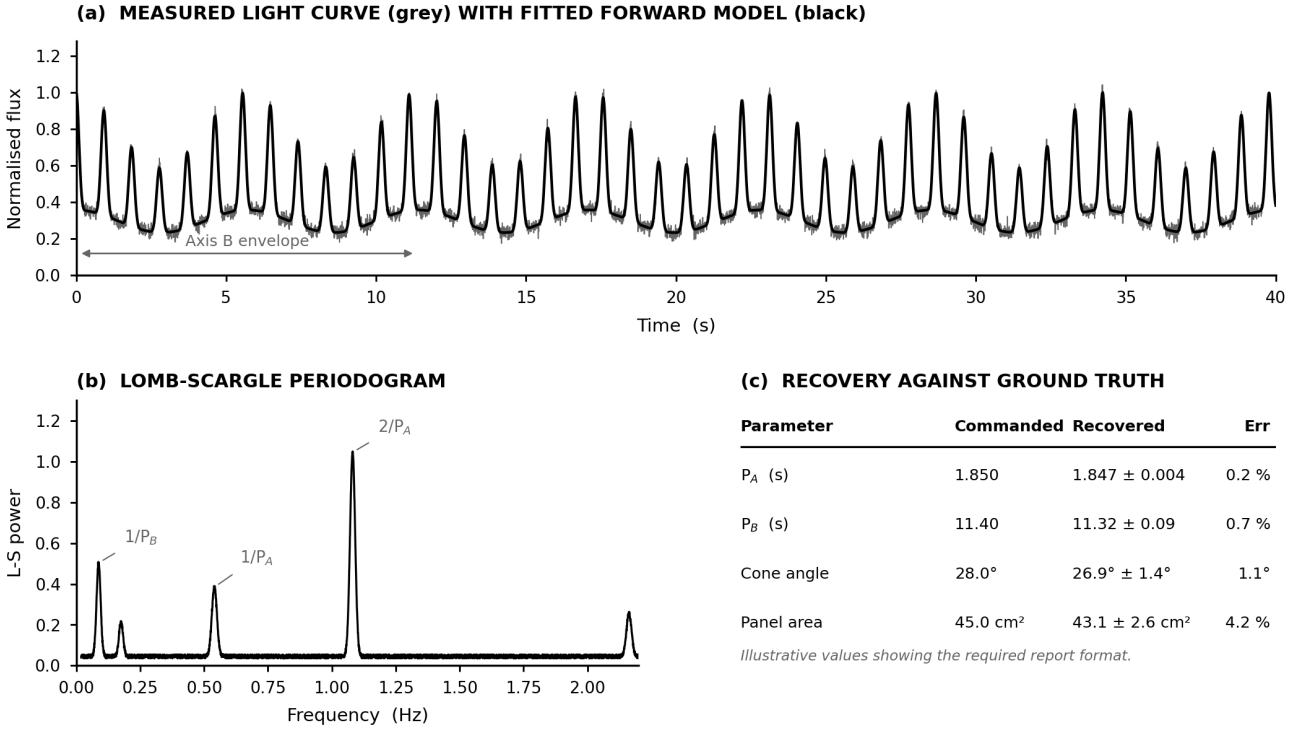
Run	Purpose	Configuration	Duration
V1	Single-axis sanity check	Model A, Axis A only at 1.85 s, Axis B held	60 s
V2	Primary validation	Model A, P_A = 1.85 s, P_B = 11.4 s	180 s
V3	Rate independence	Model A, P_A = 1.20 s, P_B = 9.7 s	180 s
V4	Shape discrimination	Model B, same rates as V2	180 s
V5	Repeatability	Model A, V2 configuration repeated after full teardown of the camera setup	180 s

V5 matters more than it appears. Repeating the primary run after deliberately disturbing and rebuilding the optical geometry tests whether the result depends on setup details that were never recorded. If V2 and V5 disagree beyond their stated error bars, the error bars are wrong — and finding that out yourself is far better than having it found for you.

SECTION 14

Expected results

The output of a successful validation run is a figure of this form, accompanied by a table comparing every recovered parameter against commanded truth.



Parameter	Commanded	Recovered	Err
P_A (s)	1.850	1.847 ± 0.004	0.2 %
P_B (s)	11.40	11.32 ± 0.09	0.7 %
Cone angle	28.0°	26.9° ± 1.4°	1.1°
Panel area	45.0 cm²	43.1 ± 2.6 cm²	4.2 %

Illustrative values showing the required report format.

FIG. 11 — Expected result format. (a) Measured light curve with the fitted forward model overlaid. (b) Lomb–Scargle periodogram showing the spin frequency, its harmonic, and the precession frequency. (c) Recovery table. Values shown are illustrative of the required reporting format, not measured data.

14.1 Acceptance criteria

Parameter	Target accuracy	Basis for the target
Axis A period	Better than 1 %	Straightforward given ≈111 frames per revolution and a 180 s baseline. Anything worse indicates a data-quality problem, not an algorithmic one.
Axis B period	Better than 3 %	Fewer cycles in the record, so intrinsically less well constrained.
Cone angle	Within 5°	Depends on envelope depth, which is sensitive to stray light and to the specular fraction.
Relative panel area	Within 15 %	Limited by residual camera non-linearity and by the simplicity of the box-wing model.
Shape discrimination	Model A and Model B distinguished at high confidence	Harmonic structure differs qualitatively between the two, so this should be robust.

A NEGATIVE RESULT IS STILL A RESULT

If the cone angle cannot be recovered to better than 15°, report that, quantify why, and identify what would fix it — more specular contrast, a longer baseline, a second viewing geometry. A well-characterised limitation is a scientific finding. An unexamined claim that everything worked is not, and will not survive questioning.

SECTION 15

Feasibility assessment

This section states plainly what is achievable in eleven days, what is not, and where the project is most likely to fail. It is written to be read by someone looking for the weak point, on the principle that it is better to identify it first.

15.1 Technical feasibility

Element	Assessment	Reasoning
Mechanism	**High confidence**	Two stepper axes on an extrusion frame with a belt reduction is routine hobby-robotics construction. No novel mechanical engineering is required.
Illumination and enclosure	**High confidence**	A DC COB LED in a light-sealed box. The only real risks are flicker and stray light, and both have explicit verification tests in Section 12.
Photometry	**High confidence**	Aperture photometry on a bright target against a black background is the easy case. Standard tooling.
Period recovery	**High confidence**	Lomb–Scargle on a high signal-to-noise periodic signal with 180 s of data at 60 fps is a well-conditioned problem.
Harmonic disambiguation	**Moderate confidence**	Requires care but is tractable, and Model B provides a diagnostic case. This is where the first genuine intellectual difficulty appears.
Axis and shape inversion	**Moderate confidence**	Convergence of a multi-parameter fit is the main risk. Mitigated by developing against synthetic data from day 1, where the true answer is known and the fit can be debugged without hardware.
Absolute area recovery	**Low confidence**	Limited by camera non-linearity and by model simplicity. Treat as a stretch objective and report relative values.

15.2 Schedule feasibility

Eleven days is tight but workable, for three specific structural reasons:

— **Nothing is imported.** Every item in Section 11 is obtainable same-day in the Delhi NCR market or within one to two days from Indian online retailers. This is the single largest schedule risk that has been designed out, and it is why no astronomy camera, no GPS module and no Raspberry Pi appears in the build.

— **Software does not wait for hardware.** Stages 5 to 8 are developed against synthetic data from day 1. A parts delay costs assembly time, not analysis time.

— **The 3D prints start on day 1** and run unattended, removing an eight-hour serial dependency from the middle of the schedule.

The honest counterpoint: there is exactly **one** contingency day. Two independent failures will consume it. The spares in Section 11 exist specifically to convert component failures from day-long setbacks into hour-long ones.

15.3 Cost feasibility

At ₹12,718 for the full build including all spares — or ₹9,098 for the lean variant — this sits well within an individual budget. Cost is not a limiting factor. Schedule and debugging time are.

15.4 Where this project will most likely fail

Ranked by probability, based on the failure modes inherent to this class of build:

1. **The inversion does not converge.** Multi-parameter fits to periodic signals have many local minima. Mitigation: synthetic data from day 1; sensible initial guesses seeded from the periodogram; fix the periods first, then fit geometry.
2. **An undiagnosed periodic artefact contaminates the light curve.** Lamp flicker, motor cogging or frame-rate beating all produce false peaks. Mitigation: DC lighting, TMC2209 drivers, and a mandatory control recording with the target stationary — any periodicity in that recording is instrumental by definition.
3. **Stray light suppresses the modulation depth.** Mitigation: the 2 % floor test in Section 12.5, performed before any data is recorded.

4. **Mechanical imbalance produces a rate error at the measured frequency.** Mitigation: the balance procedure and Checkpoint 3.
5. **A print fails late.** Mitigation: start both prints on day 1.

15.5 What would make this substantially stronger, given more time

— A second camera at a different viewing angle, permitting genuine spin-axis direction recovery rather than cone angle alone.

— A real observation of a bright satellite pass, cross-checked against published two-line elements, converting the laboratory validation into a field result.

— Full convex inversion in place of the parameterised box-wing model.

— A monochrome global-shutter camera, removing both the non-linearity and the rolling-shutter caveats — deliberately excluded here on lead-time grounds alone, not technical ones.

SECTION 16

Risk register

ID	Risk	L	I	Mitigation	Owner action
R1	Stepper driver failure	M	H	Two spare TMC2209 on the bench (A03)	Swap and continue; ≈20 min
R2	Mains-flicker artefact mistaken for signal	M	H	DC COB LED only (E01); mandatory stationary-target control recording	Inspect control recording before trusting any periodicity
R3	Motor cogging produces a false period	M	H	TMC2209 at 1/16 with StealthChop (A02)	Verify by re-running at a different microstep setting; a real signal does not move
R4	Camera auto-exposure flattens the modulation	M	H	Blackmagic Camera with locked ISO, shutter, WB; Smart HDR off	Confirm lock before every session
R5	Stray light suppresses modulation depth	M	M	Matte black interior, baffle, sealed seams	2 % floor test before recording
R6	Inversion fails to converge	M	H	Synthetic-data development from day 1; periodogram-seeded initial guesses	Fall back to period-only result, which is still a complete deliverable
R7	Cradle imbalance causes rate error at the measured frequency	M	M	Counterweight and Checkpoint 3	Re-balance; cross-check against slot-sensor period
R8	3D print failure	M	M	Prints started day 1; foam-board fallback target	Reprint or fall back
R9	Glint peak clipping	M	M	Histogram check at setup	Reduce ISO or shutter; re-record
R10	Belt skip producing a silent rate error	L	H	Correct tension at Checkpoint 2; slot sensor detects any discrepancy	The ground-truth channel makes this self-detecting
R11	Arduino failure	L	H	Spare Uno (A05)	Swap; ≈30 min including upload
R12	COB LED thermal droop	L	M	Heatsink (E03); 10 min warm-up before recording	Discard affected runs
R13	Variable frame rate in recorded video	L	H	Blackmagic Camera records constant frame rate	Verify extracted frame count equals duration × 60 exactly
R14	Two independent failures exhaust the single contingency day	M	H	Spares strategy; strict phase checkpoints	Descope to Model A period recovery only — still a defensible deliverable

L = likelihood, I = impact, each on a low / medium / high scale.

THE DESCOPE LADDER

If the schedule collapses, drop capability in this order and stop at the first rung that still fits: **(1)** Model B shape discrimination, **(2)** panel-area recovery, **(3)** cone-angle recovery, **(4)** Axis B entirely — leaving a single-axis demonstrator with validated period recovery. Rung 4 is still a complete, honest, self-validating instrument and takes about four days to build. Deciding the ladder now, while calm, is worth more than deciding it on day 9.

SECTION 17

Schedule

FIG. 12 ELEVEN-DAY SCHEDULE

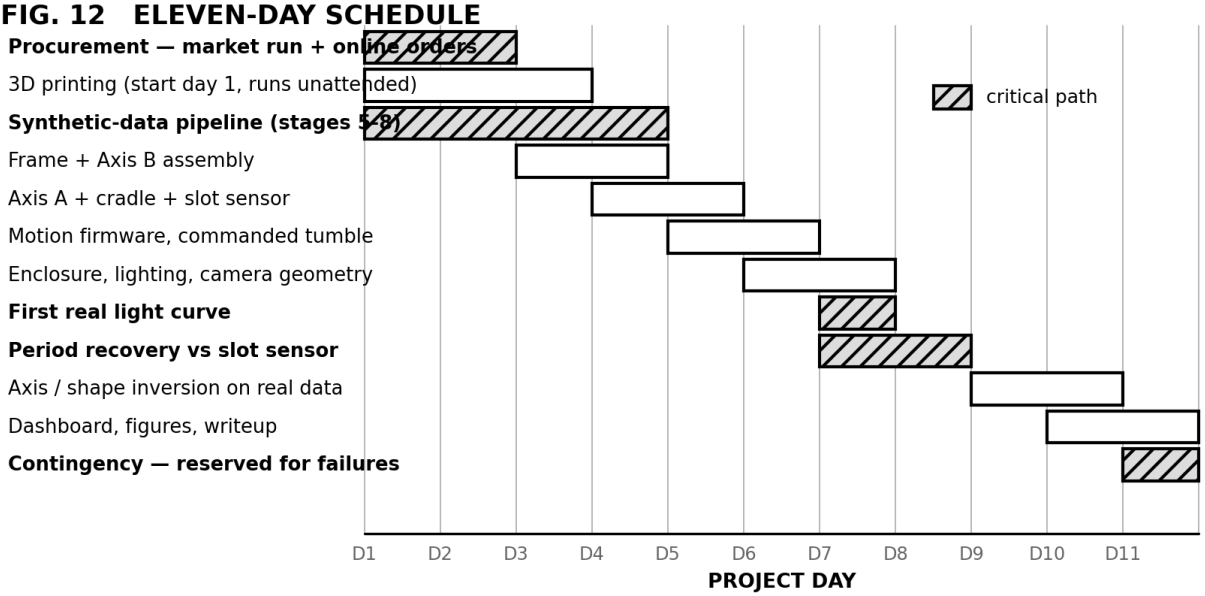


FIG. 12 — Eleven-day schedule. Hatched bars are on the critical path. Note the three parallel day-1 activities: procurement, printing and software — none of which blocks the others.

17.1 Daily detail

Day	Objective	Completion criterion
1	Procurement, print start, synthetic pipeline start	All parts ordered or bought; both prints running; synthetic light curve generated and plotted
2	Parts arrive; frame assembly; period search on synthetic data	Checkpoint 1 passed; Lomb–Scargle recovers the synthetic period
3	Axis B assembly; inversion code on synthetic data	Checkpoint 2 passed; fit converges on synthetic input
4	Axis A, cradle, slot sensor	Checkpoint 3 passed
5	Electronics, firmware, commanded tumble	Checkpoint 4 passed
6	Enclosure, lighting, camera geometry	Checkpoint 5 passed
7	First real light curve	A recorded modulation clearly visible in the frame-mean trace
8	Period recovery on real data	Recovered period agrees with slot sensor to under 1 %
9	Axis and shape inversion on real data	Fit converges; posteriors produced
10	Runs V3 to V5; figures; write-up	All validation runs complete and tabulated
11	**Contingency**	Reserved. Do not schedule new work into this day.

THE THREE THINGS TO DO IN THE FIRST TWO HOURS

1. Start both 3D prints — they run unattended and are an eight-hour serial dependency. **2.** Place online orders for the CNC shield, tripod and anything the market may not stock. **3.** Write the synthetic light-curve generator. All three are independent of each other and of everything bought later, and together they remove the three largest sources of schedule risk before the first day is over.

SECTION 18

Features and deliverables

18.1 Demonstrable features

Feature	What is shown	Robustness
Commanded two-axis tumble	The mechanism running a genuine non-principal-axis tumble at arbitrary commanded rates	Deterministic — works on demand
Live light-curve capture	Brightness trace building in real time as the target rotates	Deterministic
Independent ground truth	Slot-sensor pulse train displayed alongside the optical result	Deterministic
Period recovery with error bars	Recovered period against commanded and sensor values	High confidence
Harmonic disambiguation	Correct identification of P rather than P/2, demonstrated on both target models	Moderate to high
Spin-axis cone angle recovery	Fitted geometry with posteriors	Moderate
Shape discrimination	Model A and Model B distinguished from light curve alone	Moderate to high
Repeatability	V2 and V5 agreement across a full optical rebuild	Moderate

18.2 Deliverables at the end of the eleven days

1. The physical instrument — two-axis tumble simulator, enclosure, illumination and camera mount, operable on demand.
2. Two printed and surfaced target models with known geometry.
3. Firmware for commanded rates and timestamped ground truth.
4. A complete Python analysis pipeline from video file to fitted parameters with uncertainties.
5. A synthetic light-curve generator, usable both as a test harness and as a teaching demonstration.
6. Five validation datasets (V1–V5) with recovered-against-truth tables.
7. A results figure set in the format of Fig. 11.
8. This specification, updated to as-built status with any deviations recorded.

18.3 The claim that can honestly be made

STATEMENT OF RESULT

“We built an instrument that generates a known two-axis tumble, recorded it with an unmodified phone camera, and recovered the rotation period to better than 1 %, the precession period to better than 3 %, and the spin-axis cone angle to within a few degrees — each validated against an independent mechanical measurement rather than against our own assumptions. The same pipeline distinguishes two target geometries from their light curves alone. This is the ground-based characterisation step that active debris removal requires before any servicer can approach a derelict.”

Every number in that statement is falsifiable with equipment on the table. That is the property that makes it worth claiming.

APPENDIX A

Pin assignment and configuration reference

A.1 Controller pin map

Function	Arduino pin	Shield location	Notes
Axis A STEP	D2	X socket	Motor A — target spin
Axis A DIR	D5	X socket	—
Axis B STEP	D3	Y socket	Motor B — cradle tilt via 3:1 belt
Axis B DIR	D6	Y socket	—
Driver ENABLE	D8	Common	Active LOW. Drive HIGH to release both motors.
Slot sensor OUT	D9	X+ limit header	Polled each loop; rising edge timestamped
Sensor VCC / GND	5 V / GND	Shield power header	Do not power the sensor from the 12 V rail
Z socket	D4 / D7	Unused	Leave empty; do not fit a third driver

A.2 Microstepping jumper table — TMC2209

MS2	MS1	Microsteps	Use
LOW	LOW	1/8	Alternative if higher Axis A rates are required
LOW	HIGH	1/32	Unnecessarily fine; raises the step rate for no benefit
HIGH	LOW	1/64	Not used
HIGH	HIGH	**1/16 — SPECIFIED**	Both jumpers installed. Default for this build.

Vendor silkscreen conventions differ between TMC2209 board makers. Confirm which physical jumper positions correspond to MS1 and MS2 on the specific boards purchased before applying power.

A.3 Derived motion constants

Quantity	Value	Derivation
Full steps per motor revolution	200	1.8° step angle
Microsteps per motor revolution	3,200	200 × 16
Microsteps per Axis A revolution	3,200	Direct drive, 1:1
Microsteps per Axis B revolution	9,600	3,200 × 3 (60 T driven by 20 T)
Axis B resolution	0.0375° per microstep	360° ÷ 9,600
Axis B sweep, ±60°	3,200 microsteps	120° ÷ 0.0375°
Axis A step rate at P_A = 1.85 s	1,730 steps/s	3,200 ÷ 1.85
Axis B step rate at P_B = 11.4 s	842 steps/s	Triangular sweep: 2 × 3,200 microsteps per 11.4 s cycle

APPENDIX B

Procurement checklist

Grouped by where to buy, in the order to buy it. Take this list to the market; tick as purchased.

B.1 Physical market — Lajpat Rai Market or Nehru Place, same day

Code	Item	Specification	Qty	₹
A01	Stepper motor, NEMA 17	17HS4401 · 1.8° · 40 N·cm · 1.5 A/phase · 4-wire bipolar · 5 mm shaft	2	1,300
A02	Stepper driver, TMC2209	StepStick footprint · UART/legacy STEP-DIR · with heatsink	2	600
A03	Stepper driver, TMC2209 — SPARE	As A02	2	600
A04	Arduino Uno R3	ATmega328P · USB-B	1	450
A05	Arduino Uno R3 — SPARE	As A04	1	400
A07	Driver heatsink, adhesive	9 × 9 × 5 mm aluminium	4	60
A08	SMPS power supply	12 V · 5 A · 60 W · centre-positive barrel output	1	550
A09	DC barrel jack, female, screw-terminal	5.5 × 2.1 mm	2	50
A10	USB cable, A to B	1.5 m	1	120
B06	Ball bearing, 608ZZ	8 mm ID · 22 mm OD · 7 mm width · shielded	4	160
B07	Steel shaft, ground	8 mm dia. × 200 mm · h6 tolerance	1	120
B08	Shaft collar with grub screw	8 mm bore	2	120
C01	Aluminium extrusion, 2020	400 mm length · T-slot	2	220
C02	Aluminium extrusion, 2020	210 mm length · T-slot	2	120
C03	Aluminium extrusion, 2020	200 mm length · T-slot	2	116
C04	Corner bracket, 2020 L-type	With fixing holes, black	12	264
C05	T-nut, M5, for 2020 slot	Drop-in or slide-in	40	160
C06	Bolt, M5 × 10, socket cap	Stainless or zinc	40	120
C07	Bolt, M3 × 10, socket cap	For NEMA 17 face mounting	20	40
C08	Nut, M3	Plain	20	20
C09	Washer, M3	Plain	20	20
C12	Washer, M8, heavy	For counterweight stack	20	60
C13	Stud M8 × 40 with two nuts	Threaded rod section	1	40
C14	Rubber foot, self-adhesive	20 mm dia.	4	40
D01	IR slot / opto-interrupter module	LM393 comparator · 5 V · digital OUT · 10 mm slot	1	80
D02	IR slot sensor — SPARE	As D01	1	80
E01	COB LED module	10 W · 12 V DC direct-drive · ≈5000 K	1	250
E02	COB LED module — SPARE	As E01	1	250
E03	Aluminium heatsink	40 × 40 × 11 mm, finned	1	90
E04	Thermal adhesive pad or paste	Small tube / 20 × 20 mm pad	1	60
E05	Blackout cloth	Cotton, matte black, 2 m × 1.5 m	1	380
E07	Matte black spray paint	200 ml, acrylic, ultra-matte	1	240
F03	Spirit level, small	Bubble, adhesive or shoe-mount	1	60
G02	Aluminised mylar tape	48 mm × 10 m	1	180
G03	Blue transparent facing	Blue acrylic offcut 100 × 200 × 2 mm, or blue cellophane	1	150
G04	Grub screw, M3 × 4	Cup point	4	20
G05	Double-sided foam tape	12 mm × 5 m	1	70
H01	Hookup wire, stranded	22 AWG · 6 colours · 5 m each	1	280

Code	Item	Specification	Qty	₹
H02	Jumper wires, male-male	40-way ribbon, 20 cm	1	70
H03	Jumper wires, male-female	40-way ribbon, 20 cm	1	70
H04	Jumper wires, female-female	40-way ribbon, 20 cm	1	70
H05	JST-XH connector pairs, 4-pin	With crimp terminals	4	60
H06	Screw terminal block, 2-pin	5 mm pitch, PCB or panel	4	48
H07	Heat-shrink tubing	Assorted 2–6 mm	1	90
H08	Spiral cable wrap	6 mm × 2 m	1	60
H09	Cable tie, 100 mm	Nylon	50	40
H10	Adhesive cable clip	Small	10	30
I01	Hot glue sticks	11 mm	10	60
I02	Solder, 60/40 rosin core	0.8 mm · 50 g	1	150
I03	Electrical tape	PVC, black	2	40
I04	Masking tape	24 mm	1	40
I05	Sandpaper	400 and 800 grit	2	30
I06	Isopropyl alcohol	99 % · 100 ml	1	90
I07	Threadlocker, medium strength	Blue · 5 ml	1	120
I08	Lint-free cloth / cotton buds	Pack	1	50

Market subtotal: ₹9,058

B.2 Online — order on day 1

Code	Item	Specification	Qty	₹	Source
A06	CNC Shield V3	4 × StepStick sockets · screw terminal · microstep jumpers	1	300	Amazon / Robu
B01	GT2 timing pulley, 20 T	5 mm bore · 6 mm belt width · with grub screws	1	120	Amazon / Robu
B02	GT2 timing pulley, 60 T	8 mm bore · 6 mm belt width	1	280	Amazon / Robu
B03	GT2 belt, closed loop	280 mm (140 T) · 6 mm wide	1	150	Amazon / Robu
B04	GT2 belt — SPARE	300 mm (150 T) · 6 mm wide	1	150	Amazon / Robu
B05	Flexible shaft coupler	5 mm to 8 mm · aluminium, helical cut	1	150	Amazon / Robu
B09	Aluminium hub, grub-screw	5 mm bore · M3 grub · flange for target mounting	1	180	Amazon / Robu
C10	NEMA 17 mounting bracket, L-type	Steel, slotted for belt tensioning	2	240	Amazon / Robu
F01	Tripod	≈1.3 m · pan-tilt head · 1/4-20 thread	1	700	Amazon / Market
F02	Phone clamp	Universal, 1/4-20 socket	1	200	Amazon / Market
G01	PLA filament	1.75 mm · white · 1 kg spool	1	900	Amazon / Robu

Online subtotal: ₹3,370 — order these first; they are the only items with any lead time.

B.3 Other sources

Code	Item	Specification	Qty	₹	Source
C11	Cradle and bearing housings	3D printed PETG, or 2 mm aluminium sheet	1	0	Printed
D03	Encoder tab	15 × 10 × 1 mm, opaque	1	0	Printed
E06	Black chart paper	A2, matte	10	150	Stationers
E08	Foam board	5 mm, A3	2	120	Stationers
E09	Grayscale step wedge	Printed reflectance ladder on matte paper	1	20	Print shop
E10	Cardboard carton	≈600 × 450 × 450 mm	1	0	Salvage

Reconciliation: market ₹9,058 + online ₹3,370 + other ₹290 = **₹12,718**, equal to the Section 11 total. Each item appears in exactly one list; where two vendors are shown in the BOM, the first is treated as primary.

B.4 Tools assumed available

Not costed in Section 11. Confirm access before day 1 — a missing tool on day 3 costs the same as a missing part.

Tool	Use
Digital multimeter	Coil-pair identification; driver V_{REF} setting. **Mandatory** — do not start without one.
Soldering iron, 25–40 W, with solder and flux	All permanent joints
Hot glue gun	Target assembly, cable clipping
Hex key set, 1.5–6 mm	Extrusion bolts, grub screws, pulleys
Screwdriver set, small	Terminal blocks, driver trimpots
Wire cutter and stripper	Harness fabrication
Steel rule 300 mm and digital caliper	Frame setting-out; shaft and bore checks
Hacksaw with fine blade, or shop cutting	Extrusion to length — shop cutting is faster and squarer
Spirit level, small	Post plumb and camera levelling
3D printer with PLA capability	Target models, cradle, bearing housings

APPENDIX C

Troubleshooting guide

Symptom	Most likely cause	Action
Motor buzzes or vibrates but does not turn	Coil pairs mis-wired	Power down. Re-identify pairs with a multimeter (2–3 Ω within a pair). Rewire.
Motor turns in the wrong direction	Coil polarity	Reverse one pair only, or invert DIR in firmware.
Driver becomes very hot within a minute	V_{REF} set too high	Power down, reduce V_{REF} toward 0.5 V. Confirm the heatsink is fitted.
Motion is rough or steps are lost	Belt over-tensioned, or mechanical binding	Release belt tension and turn each axis by hand. Binding must be cured mechanically, not by raising current.
No pulses from the slot sensor	Tab misaligned, or sensor powered from the wrong rail	Confirm the tab passes through the slot. Confirm the module is on 5 V, not 12 V. Check the module's own LED indicator.
Pulse intervals are erratic	Electrical noise on the sensor line, or a loose tab	Shorten and route the sensor lead away from motor cables. Threadlock the tab fixing.
Light curve shows a strong period matching no commanded rate	Lamp flicker, motor cogging, or a frame-rate beat	Record the control run with the target stationary. Any periodicity there is instrumental. Then re-run at a different microstep setting — an instrumental peak moves, a real one does not.
Modulation depth is much smaller than expected	Stray light, or insufficient specular contrast on the target	Run the 2 % stray-light floor test. If it passes, add or refresh the panel facing.
Light curve drifts slowly downward over a recording	LED thermal droop	Confirm the heatsink bond. Allow 10 minutes of warm-up before recording.
Glint peaks are flat-topped	Sensor clipping	Reduce ISO or shorten shutter. Re-record — clipped data cannot be recovered.
Brightness varies with no target motion	Camera auto-exposure still active	Re-lock ISO, shutter and white balance. Confirm Smart HDR is off in iOS Settings.
Extracted frame count does not equal duration × 60	Variable frame rate recording	Re-record in Blackmagic Camera with a fixed frame rate. Do not attempt to correct this in post-processing.
Periodogram shows a strong peak at twice the expected frequency	Harmonic ambiguity — expected behaviour for a symmetric target	This is not a fault. Apply the phase-folding test in pipeline stage 6.
Inversion converges to different answers from different starting points	Local minima in the parameter space	Fix the periods from the periodogram first, then fit geometry with those held. Seed from the synthetic-data solution.
Recovered period disagrees with the slot sensor by a factor of exactly 2	Harmonic ambiguity not resolved	Stage 6 of the pipeline has been skipped or has failed. The slot sensor is correct.

END OF DOCUMENT — SDC-001 REV A